

# KIT SAFETY DATA SHEET



Kit Product Name Syphilis Total Ab, 96 Tests

Kit Catalogue Number(s) 72530

Revision date 31-Jan-2025

## Kit Contents

| Catalogue Number(s)        | Product Name                                    |
|----------------------------|-------------------------------------------------|
| 7360G, 5180S, 7435S        | R8 - Substrat Buffer, 60 mL                     |
| 7360J, 5180U, 7361H, 7337Z | R10 - Stopping Solution, 28 mL                  |
| 7361A, 7360S, 7360Z        | R2 - Concentrated washing solution (20X), 70 mL |
| 7436L, 7436H               | R9 - Chromogen: TMB Solution (11X), 5 mL        |
| 7253A                      | R1 - Microplate, 12 strips x 8 wells            |
| 7253F                      | R3 - Negative control, 2.1 ml                   |
| 7253K                      | R4 - Positive Control, 1.6 ml                   |
| 7253L                      | R6 - Conjugate, 8 ml                            |
| 7253H                      | R7 - Conjugate 2, 14 ml                         |



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
UK REACH Regulations (SI 2019/758 as amended)

Revision date 09-Aug-2024

Revision Number 2.3

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### 1.1. Product identifier

**Product Name** R8 - Substrat Buffer, 60 mL  
**Catalogue Number(s)** 7360G, 5180S, 7435S, 7337V  
**Pure substance/mixture** Mixture

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

**Recommended use** In vitro diagnostic Restricted to professional users  
**Uses advised against** No information available

### 1.3. Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories Ltd  
The Junction  
Station Road  
Watford, WD17 1ET  
UK

For further information, please contact

**Technical Service** 00800 00246 723  
Techsupport.UK@bio-rad.com

### 1.4. Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC UK: 44-870-8200418

## SECTION 2: Hazards identification

### 2.1. Classification of the substance or mixture

Not classified Not a hazardous substance or mixture according to the Globally Harmonised System (GHS)

### 2.2. Label elements

Not classified Not a hazardous substance or mixture according to the Globally Harmonised System (GHS)

#### **Hazard statements**

Not classified Not a hazardous substance or mixture according to the Globally Harmonised System (GHS)

### 2.3. Other hazards

No information available.

## SECTION 3: Composition/information on ingredients

**3.1 Substances**

Not applicable

**3.2 Mixtures**

| Chemical name          | Weight-% | EC No (EU Index No)         | UK REACH registration number | Classification according to GB CLP (SI 2020/1567 as amended) | Specific concentration limit (SCL) | M-Factor | M-Factor (long-term) |
|------------------------|----------|-----------------------------|------------------------------|--------------------------------------------------------------|------------------------------------|----------|----------------------|
| Citric acid<br>77-92-9 | 1.58     | 201-069-1<br>(607-750-00-3) | -                            | Eye Irrit. 2<br>(H319)<br>STOT SE 3<br>(H335)                | -                                  | -        | -                    |

**Full text of H- and EUH-phrases: see section 16**This product does not contain candidate substances of very high concern at a concentration  $\geq 0.1\%$  (UK REACH Article 59)**SECTION 4: First aid measures****4.1. Description of first aid measures**

|              |                                                                                                                   |
|--------------|-------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                              |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a doctor. |
| Skin contact | Wash skin with soap and water. In the case of skin irritation or allergic reactions see a doctor.                 |
| Ingestion    | Rinse mouth.                                                                                                      |

**4.2. Most important symptoms and effects, both acute and delayed**

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

**4.3. Indication of any immediate medical attention and special treatment needed**

|                 |                        |
|-----------------|------------------------|
| Note to doctors | Treat symptomatically. |
|-----------------|------------------------|

**SECTION 5: Firefighting measures****5.1. Extinguishing media**

|                                |                                                                                                         |
|--------------------------------|---------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
| Large Fire                     | CAUTION: Use of water spray when fighting fire may be inefficient.                                      |
| Unsuitable extinguishing media | Do not scatter spilled material with high pressure water streams.                                       |

**5.2. Special hazards arising from the substance or mixture**

|                                            |                           |
|--------------------------------------------|---------------------------|
| Specific hazards arising from the chemical | No information available. |
|--------------------------------------------|---------------------------|

**5.3. Advice for firefighters**

|                                                                       |                                                                                                                                    |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Special protective equipment and precautions for fire-fighters</b> | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|

## SECTION 6: Accidental release measures

### 6.1. Personal precautions, protective equipment and emergency procedures

|                                 |                                                   |
|---------------------------------|---------------------------------------------------|
| <b>Personal precautions</b>     | Ensure adequate ventilation.                      |
| <b>For emergency responders</b> | Use personal protection recommended in Section 8. |

### 6.2. Environmental precautions

|                                  |                                                       |
|----------------------------------|-------------------------------------------------------|
| <b>Environmental precautions</b> | See Section 12 for additional Ecological Information. |
|----------------------------------|-------------------------------------------------------|

### 6.3. Methods and material for containment and cleaning up

|                                        |                                                                                      |
|----------------------------------------|--------------------------------------------------------------------------------------|
| <b>Methods for containment</b>         | Prevent further leakage or spillage if safe to do so.                                |
| <b>Methods for cleaning up</b>         | Take up mechanically, placing in appropriate containers for disposal.                |
| <b>Prevention of secondary hazards</b> | Clean contaminated objects and areas thoroughly observing environmental regulations. |

### 6.4. Reference to other sections

|                                    |                                                                          |
|------------------------------------|--------------------------------------------------------------------------|
| <b>Reference to other sections</b> | See section 8 for more information. See section 13 for more information. |
|------------------------------------|--------------------------------------------------------------------------|

## SECTION 7: Handling and storage

### 7.1. Precautions for safe handling

|                                       |                                                                      |
|---------------------------------------|----------------------------------------------------------------------|
| <b>Advice on safe handling</b>        | Ensure adequate ventilation.                                         |
| <b>General hygiene considerations</b> | Wash hands before breaks and immediately after handling the product. |

### 7.2. Conditions for safe storage, including any incompatibilities

|                           |                                                    |
|---------------------------|----------------------------------------------------|
| <b>Storage Conditions</b> | Store according to product and label instructions. |
|---------------------------|----------------------------------------------------|

### 7.3. Specific end use(s)

|                                      |                                                                  |
|--------------------------------------|------------------------------------------------------------------|
| <b>Risk Management Methods (RMM)</b> | The information required is contained in this Safety Data Sheet. |
|--------------------------------------|------------------------------------------------------------------|

## SECTION 8: Exposure controls/personal protection

### 8.1. Control parameters

|                                                |                                                                                                                                                             |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Limits</b>                         | This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies. |
| <b>Biological occupational exposure limits</b> | This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies.            |
| <b>Derived No Effect Level (DNEL)</b>          | No information available.                                                                                                                                   |

**Predicted No Effect Concentration (PNEC)** No information available.

## 8.2. Exposure controls

**Engineering controls** No information available.

## Personal protective equipment

**Eye/face protection** No special protective equipment required.

**Skin and body protection** No special protective equipment required.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

# SECTION 9: Physical and chemical properties

## 9.1. Information on basic physical and chemical properties

|                        |                          |
|------------------------|--------------------------|
| <b>Physical state</b>  | Liquid                   |
| <b>Appearance</b>      | aqueous solution         |
| <b>Colour</b>          | colourless               |
| <b>Odour</b>           | Odourless.               |
| <b>Odour threshold</b> | No information available |

| <u>Property</u>                                | <u>Values</u>            | <u>Remarks • Method</u> |
|------------------------------------------------|--------------------------|-------------------------|
| <b>Melting point / freezing point</b>          | No data available        | None known              |
| <b>Initial boiling point and boiling range</b> | No data available        | None known              |
| <b>Flammability</b>                            | No data available        | None known              |
| <b>Flammability Limit in Air</b>               |                          | None known              |
| <b>Upper flammability or explosive limits</b>  | No data available        |                         |
| <b>Lower flammability or explosive limits</b>  | No data available        |                         |
| <b>Flash point</b>                             | No data available        | None known              |
| <b>Autoignition temperature</b>                | No data available        |                         |
| <b>Decomposition temperature</b>               |                          | None known              |
| <b>pH</b>                                      | 4                        |                         |
| <b>pH (as aqueous solution)</b>                | No data available        | None known              |
| <b>Kinematic viscosity</b>                     | No data available        | None known              |
| <b>Dynamic viscosity</b>                       | No data available        | None known              |
| <b>Water solubility</b>                        | Miscible in water        |                         |
| <b>Solubility(ies)</b>                         | No data available        | None known              |
| <b>Partition coefficient</b>                   | No data available        | None known              |
| <b>Vapour pressure</b>                         | No data available        | None known              |
| <b>Relative density</b>                        | No data available        | None known              |
| <b>Bulk density</b>                            | No data available        |                         |
| <b>Liquid Density</b>                          | No data available        |                         |
| <b>Relative vapour density</b>                 | No data available        | None known              |
| <b>Particle characteristics</b>                |                          |                         |
| <b>Particle Size</b>                           | No information available |                         |
| <b>Particle Size Distribution</b>              | No information available |                         |

## 9.2. Other information

# SECTION 10: Stability and reactivity

## 10.1. Reactivity

Reactivity No information available.

#### 10.2. Chemical stability

Stability Stable under normal conditions.

#### Explosion data

Sensitivity to mechanical impact None.

Sensitivity to static discharge None.

#### 10.3. Possibility of hazardous reactions

Possibility of hazardous reactions None under normal processing.

#### 10.4. Conditions to avoid

Conditions to avoid None known based on information supplied.

#### 10.5. Incompatible materials

Incompatible materials None known based on information supplied.

#### 10.6. Hazardous decomposition products

Hazardous decomposition products None known based on information supplied.

### SECTION 11: Toxicological information

#### 11.1. Information on toxicological effects

##### Information on likely routes of exposure

##### Product Information

Inhalation Specific test data for the substance or mixture is not available.

Eye contact Specific test data for the substance or mixture is not available.

Skin contact Specific test data for the substance or mixture is not available.

Ingestion Specific test data for the substance or mixture is not available.

##### Symptoms related to the physical, chemical and toxicological characteristics

Symptoms No information available.

##### Acute toxicity

##### Numerical measures of toxicity

The following values are calculated based on chapter 3.1 of the GHS document

ATEmix (oral) 145,197.40 mg/kg

ATEmix (dermal) 406,504.10 mg/kg

ATEmix (inhalation-dust/mist) 411.80 mg/l

##### Component Information

| Chemical name | Oral LD50        | Dermal LD50          | Inhalation LC50 |
|---------------|------------------|----------------------|-----------------|
| Citric acid   | = 3 g/kg ( Rat ) | > 2000 mg/kg ( Rat ) | -               |

##### Delayed and immediate effects as well as chronic effects from short and long-term exposure

Skin corrosion/irritation No information available.

Serious eye damage/eye irritation No information available.

Respiratory or skin sensitisation No information available.

Germ cell mutagenicity No information available.

Carcinogenicity No information available.

Reproductive toxicity No information available.

STOT - single exposure No information available.

STOT - repeated exposure No information available.

Aspiration hazard No information available.

Other adverse effects No information available.

## SECTION 12: Ecological information

### 12.1. Toxicity

**Ecotoxicity** The environmental impact of this product has not been fully investigated.

| Chemical name | Algae/aquatic plants | Fish                                               | Toxicity to microorganisms | Crustacea |
|---------------|----------------------|----------------------------------------------------|----------------------------|-----------|
| Citric acid   | -                    | LC50: =1516mg/L (96h, <i>Lepomis macrochirus</i> ) | -                          | -         |

### 12.2. Persistence and degradability

**Persistence and degradability** No information available.

### 12.3. Bioaccumulative potential

**Bioaccumulation**

#### Component Information

| Chemical name | Partition coefficient |
|---------------|-----------------------|
| Citric acid   | -1.72                 |

### 12.4. Mobility in soil

**Mobility in soil** No information available.

### 12.5. Results of PBT and vPvB assessment

**PBT and vPvB assessment** No information available.

| Chemical name | PBT and vPvB assessment         |
|---------------|---------------------------------|
| Citric acid   | The substance is not PBT / vPvB |

**12.6. Endocrine disrupting properties**

No information available.

## SECTION 13: Disposal considerations

**13.1. Waste treatment methods**

|                                            |                                                                                                                 |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Waste from residues/unused products</b> | Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. |
| <b>Contaminated packaging</b>              | Do not reuse empty containers.                                                                                  |

## SECTION 14: Transport information

**IATA**

|                                          |                |
|------------------------------------------|----------------|
| <b>14.1 UN number or ID number</b>       | Not regulated  |
| <b>14.2 UN proper shipping name</b>      | Not regulated  |
| <b>14.3 Transport hazard class(es)</b>   | Not regulated  |
| <b>14.4 Packing group</b>                | Not regulated  |
| <b>14.5 Environmental hazards</b>        | Not applicable |
| <b>14.6 Special precautions for user</b> |                |
| <b>Special Provisions</b>                | None           |

**IMDG**

|                                                                     |                          |
|---------------------------------------------------------------------|--------------------------|
| <b>14.1 UN number or ID number</b>                                  | Not regulated            |
| <b>14.2 UN proper shipping name</b>                                 | Not regulated            |
| <b>14.3 Transport hazard class(es)</b>                              | Not regulated            |
| <b>14.4 Packing group</b>                                           | Not regulated            |
| <b>14.5 Environmental hazards</b>                                   | Not applicable           |
| <b>14.6 Special precautions for user</b>                            |                          |
| <b>Special Provisions</b>                                           | None                     |
| <b>14.7 Maritime transport in bulk according to IMO instruments</b> | No information available |

**RID**

|                                          |                |
|------------------------------------------|----------------|
| <b>14.1 UN number or ID number</b>       | Not regulated  |
| <b>14.2 UN proper shipping name</b>      | Not regulated  |
| <b>14.3 Transport hazard class(es)</b>   | Not regulated  |
| <b>14.4 Packing group</b>                | Not regulated  |
| <b>14.5 Environmental hazards</b>        | Not applicable |
| <b>14.6 Special precautions for user</b> |                |
| <b>Special Provisions</b>                | None           |

**ADR**

|                                          |                |
|------------------------------------------|----------------|
| <b>14.1 UN number or ID number</b>       | Not regulated  |
| <b>14.2 UN proper shipping name</b>      | Not regulated  |
| <b>14.3 Transport hazard class(es)</b>   | Not regulated  |
| <b>14.4 Packing group</b>                | Not regulated  |
| <b>14.5 Environmental hazards</b>        | Not applicable |
| <b>14.6 Special precautions for user</b> |                |
| <b>Special Provisions</b>                | None           |

## SECTION 15: Regulatory information

**15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture****National regulations****Authorisations and/or restrictions on use:**

This product does not contain substances subject to authorisation (UK REACH - Annex XIV). This product does not contain



substances subject to restriction (UK REACH - Annex XVII).

#### Persistent Organic Pollutants

Not applicable

#### Export Notification requirements

Not applicable

#### Named dangerous substances per COMAH (SI 2015/483 as amended)

Not applicable

#### The Ozone-Depleting Substances Regulations 2015

Not applicable

#### The Biocidal Products Regulations 2001 (as amended)

| Chemical name         | The Biocidal Products Regulations 2001 (as amended) |
|-----------------------|-----------------------------------------------------|
| Citric acid - 77-92-9 | PT2                                                 |

#### The Water Environment (Water Framework Directive) (England and Wales) Regulations 2017 (as amended)

Not applicable

#### Poisons and Explosive Precursors

Not applicable

#### International Inventories

Contact supplier for inventory compliance status

#### 15.2. Chemical safety assessment

#### Chemical Safety Report

No information available

### SECTION 16: Other information

#### Key or legend to abbreviations and acronyms used in the safety data sheet

#### Full text of any hazard and/or precautionary statements referred to under Sections 2-15

H319 - Causes serious eye irritation

H335 - May cause respiratory irritation

#### Legend

SVHC: Substances of Very High Concern for Authorisation:

#### Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | Sk*  | Skin designation                 |
| +       | Sensitisers                 |      |                                  |

#### Classification procedure

Classification according to Regulation (EC) No. 1272/2008 [CLP]

|                                       | Method Used        |
|---------------------------------------|--------------------|
| Acute oral toxicity                   | Calculation method |
| Acute dermal toxicity                 | Calculation method |
| Acute inhalation toxicity - gas       | Calculation method |
| Acute inhalation toxicity - vapour    | Calculation method |
| Acute inhalation toxicity - dust/mist | Calculation method |
| Skin corrosion/irritation             | Calculation method |
| Serious eye damage/eye irritation     | Calculation method |
| Respiratory sensitisation             | Calculation method |
| Skin sensitisation                    | Calculation method |
| Mutagenicity                          | Calculation method |
| Carcinogenicity                       | Calculation method |
| Reproductive toxicity                 | Calculation method |
| STOT - single exposure                | Calculation method |
| STOT - repeated exposure              | Calculation method |

Acute aquatic toxicity  
Chronic aquatic toxicity  
Aspiration hazard  
Ozone

Calculation method  
Calculation method  
Calculation method  
Calculation method

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
European Chemicals Agency (ECHA) Committee for Risk Assessment (ECHA\_RAC)  
European Chemicals Agency (ECHA) (ECHA\_API)  
Environmental Protection Agency  
Acute Exposure Guideline Level(s) (AEGl(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
National Institute of Technology and Evaluation (NITE)  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
U.S. National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
World Health Organization

**Revision date** 09-Aug-2024

**Revision Note** Significant changes throughout SDS. Review all sections

**This SDS complies with the requirements of UK REACH Regulations SI 2019/758 (as amended)**

**Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work**

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**

**UK SDS version information - XGHS**

UL release:  
GHS Revision 7  
2024 Q4

Full text of any hazard and/or precautionary H319 - Causes serious eye irritation H335 - May cause respiratory irritation statements referred to under Sections 2-15

| Chemical name | Classification according to GB CLP (SI 2020/1567 as amended) | Specific concentration limit (SCL) |
|---------------|--------------------------------------------------------------|------------------------------------|
| Citric acid   | Eye Irrit. 2 (H319)<br>STOT SE 3 (H335)                      |                                    |



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
UK REACH Regulations (SI 2019/758 as amended)

Revision date 09-Aug-2024

Revision Number 3.2

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### 1.1. Product identifier

**Product Name** R10 - Stopping Solution, 28 mL

**Catalogue Number(s)** 7360J, 5180U, 7361H, 7337Z

**Pure substance/mixture** Mixture

Contains Sulfuric acid

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

**Recommended use** In vitro diagnostic Restricted to professional users

**Uses advised against** No information available

### 1.3. Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories Ltd  
The Junction  
Station Road  
Watford, WD17 1ET  
UK

For further information, please contact

**Technical Service** 00800 00246 723  
Techsupport.UK@bio-rad.com

### 1.4. Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC UK: 44-870-8200418

## SECTION 2: Hazards identification

### 2.1. Classification of the substance or mixture

|                                   |                     |
|-----------------------------------|---------------------|
| Skin corrosion/irritation         | Category 1 - (H314) |
| Serious eye damage/eye irritation | Category 1 - (H318) |

### 2.2. Label elements

Contains Sulfuric acid



**Signal word**

Danger

**Hazard statements**

H314 - Causes severe skin burns and eye damage

**Precautionary statements**

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
 P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
 P301 + P330 + P331 - IF SWALLOWED: Rinse mouth. Do NOT induce vomiting  
 P501 - Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable  
 P280 - Wear protective gloves, protective clothing, eye protection and face protection

**2.3. Other hazards**

No information available.

**SECTION 3: Composition/information on ingredients****3.1 Substances**

Not applicable

**3.2 Mixtures**

| Chemical name              | Weight-% | EC No (EU Index No)         | UK REACH registration number | Classification according to GB CLP (SI 2020/1567 as amended) | Specific concentration limit (SCL)                                                       | M-Factor | M-Factor (long-term) |
|----------------------------|----------|-----------------------------|------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------------|----------|----------------------|
| Sulfuric acid<br>7664-93-9 | 4.8      | 231-639-5<br>(016-020-00-8) | -                            | Skin Corr. 1A<br>(H314)<br>Eye Dam. 1<br>(H318)              | Eye Irrit. 2 ::<br>5%≤C<15%<br>Skin Corr. 1A ::<br>C≥15%<br>Skin Irrit. 2 ::<br>5%≤C<15% | -        | -                    |

**Full text of H- and EUH-phrases: see section 16**

This product does not contain candidate substances of very high concern at a concentration ≥ 0.1% (UK REACH Article 59)

**SECTION 4: First aid measures****4.1. Description of first aid measures****General advice**

Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.

**Inhalation**

Remove to fresh air. If breathing has stopped, give artificial respiration. Get medical attention immediately. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. If breathing is difficult, (trained personnel should) give oxygen. Delayed pulmonary edema may occur.

**Eye contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Keep eye wide open while rinsing. Do not rub affected area. Remove contact lenses, if present and easy to do. Continue rinsing. Get immediate medical attention.

**Skin contact**

Wash off immediately with soap and plenty of water while removing all contaminated clothes

and shoes. Get immediate medical attention.

**Ingestion** Do NOT induce vomiting. Rinse mouth. Never give anything by mouth to an unconscious person. Get immediate medical attention.

**Self-protection of the first aider** Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. Avoid contact with skin, eyes or clothing. Avoid direct contact with skin. Use barrier to give mouth-to-mouth resuscitation. Wear personal protective clothing (see section 8).

#### **4.2. Most important symptoms and effects, both acute and delayed**

**Symptoms** Burning sensation.

#### **4.3. Indication of any immediate medical attention and special treatment needed**

**Note to doctors** Product is a corrosive material. Use of gastric lavage or emesis is contra-indicated. Possible perforation of stomach or esophagus should be investigated. Do not give chemical antidotes. Asphyxia from glottal edema may occur. Marked decrease in blood pressure may occur with moist rales, frothy sputum, and high pulse pressure.

### **SECTION 5: Firefighting measures**

#### **5.1. Extinguishing media**

**Suitable Extinguishing Media** Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Large Fire** CAUTION: Use of water spray when fighting fire may be inefficient.

**Unsuitable extinguishing media** Do not scatter spilled material with high pressure water streams.

#### **5.2. Special hazards arising from the substance or mixture**

**Specific hazards arising from the chemical** The product causes burns of eyes, skin and mucous membranes. Thermal decomposition can lead to release of irritating gases and vapours.

#### **5.3. Advice for firefighters**

**Special protective equipment and precautions for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

### **SECTION 6: Accidental release measures**

#### **6.1. Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Attention! Corrosive material. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

**Other information** Refer to protective measures listed in Sections 7 and 8.

**For emergency responders** Use personal protection recommended in Section 8.

#### **6.2. Environmental precautions**

**Environmental precautions** Prevent further leakage or spillage if safe to do so. Should not be released into the environment. Do not allow to enter into soil/subsoil. Prevent product from entering drains.

#### **6.3. Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

|                                                |                                                                                      |
|------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Methods for cleaning up</b>                 | Take up mechanically, placing in appropriate containers for disposal.                |
| <b>Prevention of secondary hazards</b>         | Clean contaminated objects and areas thoroughly observing environmental regulations. |
| <b><u>6.4. Reference to other sections</u></b> |                                                                                      |
| <b>Reference to other sections</b>             | See section 8 for more information. See section 13 for more information.             |

## SECTION 7: Handling and storage

### 7.1. Precautions for safe handling

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Advice on safe handling</b>        | Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. In case of insufficient ventilation, wear suitable respiratory equipment. Handle product only in closed system or provide appropriate exhaust ventilation. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash it before reuse.                                                             |
| <b>General hygiene considerations</b> | Avoid contact with skin, eyes or clothing. Wear suitable gloves and eye/face protection. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Contaminated work clothing should not be allowed out of the workplace. Regular cleaning of equipment, work area and clothing is recommended. Wash hands before breaks and immediately after handling the product. |

### 7.2. Conditions for safe storage, including any incompatibilities

|                           |                                                                                                                                                                                                                                         |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Storage Conditions</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Protect from moisture. Store locked up. Keep out of the reach of children. Store away from other materials. Store according to product and label instructions. |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.3. Specific end use(s)

|                                      |                                                                  |
|--------------------------------------|------------------------------------------------------------------|
| <b>Risk Management Methods (RMM)</b> | The information required is contained in this Safety Data Sheet. |
|--------------------------------------|------------------------------------------------------------------|

## SECTION 8: Exposure controls/personal protection

### 8.1. Control parameters

#### Exposure Limits

| Chemical name              | United Kingdom                                              |
|----------------------------|-------------------------------------------------------------|
| Sulfuric acid<br>7664-93-9 | TWA: 0.05 mg/m <sup>3</sup><br>STEL: 0.15 mg/m <sup>3</sup> |

|                                                |                                                                                                                                                  |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Biological occupational exposure limits</b> | This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies. |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|

|                                       |                           |
|---------------------------------------|---------------------------|
| <b>Derived No Effect Level (DNEL)</b> | No information available. |
|---------------------------------------|---------------------------|

|                                                 |                           |
|-------------------------------------------------|---------------------------|
| <b>Predicted No Effect Concentration (PNEC)</b> | No information available. |
|-------------------------------------------------|---------------------------|

### 8.2. Exposure controls

|                             |                           |
|-----------------------------|---------------------------|
| <b>Engineering controls</b> | No information available. |
|-----------------------------|---------------------------|

#### Personal protective equipment

|                            |                                                       |
|----------------------------|-------------------------------------------------------|
| <b>Eye/face protection</b> | Tight sealing safety goggles. Face protection shield. |
|----------------------------|-------------------------------------------------------|

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hand protection</b>                | Wear suitable gloves. Impervious gloves.                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Skin and body protection</b>       | Wear suitable protective clothing. Long sleeved clothing. Chemical resistant apron.                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Respiratory protection</b>         | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.                                                                                                                                                                                                                                                                              |
| <b>General hygiene considerations</b> | Avoid contact with skin, eyes or clothing. Wear suitable gloves and eye/face protection. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Contaminated work clothing should not be allowed out of the workplace. Regular cleaning of equipment, work area and clothing is recommended. Wash hands before breaks and immediately after handling the product. |

## SECTION 9: Physical and chemical properties

### 9.1. Information on basic physical and chemical properties

|                        |                          |
|------------------------|--------------------------|
| <b>Physical state</b>  | Liquid                   |
| <b>Appearance</b>      | aqueous solution         |
| <b>Colour</b>          | colourless               |
| <b>Odour</b>           | Low.                     |
| <b>Odour threshold</b> | No information available |

| <u>Property</u>                                | <u>Values</u>            | <u>Remarks • Method</u> |
|------------------------------------------------|--------------------------|-------------------------|
| <b>Melting point / freezing point</b>          | No data available        | None known              |
| <b>Initial boiling point and boiling range</b> | No data available        | None known              |
| <b>Flammability</b>                            | No data available        | None known              |
| <b>Flammability Limit in Air</b>               |                          | None known              |
| <b>Upper flammability or explosive limits</b>  | No data available        |                         |
| <b>Lower flammability or explosive limits</b>  | No data available        |                         |
| <b>Flash point</b>                             | No data available        | None known              |
| <b>Autoignition temperature</b>                | No data available        | None known              |
| <b>Decomposition temperature</b>               |                          | None known              |
| <b>pH</b>                                      | < 2                      |                         |
| <b>pH (as aqueous solution)</b>                | No data available        | None known              |
| <b>Kinematic viscosity</b>                     | No data available        | None known              |
| <b>Dynamic viscosity</b>                       | No data available        | None known              |
| <b>Water solubility</b>                        | Miscible in water        |                         |
| <b>Solubility(ies)</b>                         | No data available        | None known              |
| <b>Partition coefficient</b>                   | No data available        | None known              |
| <b>Vapour pressure</b>                         | No data available        | None known              |
| <b>Relative density</b>                        | No data available        | None known              |
| <b>Bulk density</b>                            | No data available        |                         |
| <b>Liquid Density</b>                          | No data available        |                         |
| <b>Relative vapour density</b>                 | No data available        | None known              |
| <b>Particle characteristics</b>                |                          |                         |
| <b>Particle Size</b>                           | No information available |                         |
| <b>Particle Size Distribution</b>              | No information available |                         |

### 9.2. Other information

## SECTION 10: Stability and reactivity

### 10.1. Reactivity

|                   |                           |
|-------------------|---------------------------|
| <b>Reactivity</b> | No information available. |
|-------------------|---------------------------|

### 10.2. Chemical stability

**Stability** Stable under normal conditions.

**Explosion data**

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

**10.3. Possibility of hazardous reactions**

**Possibility of hazardous reactions** None under normal processing.

**10.4. Conditions to avoid**

**Conditions to avoid** Exposure to air or moisture over prolonged periods.

**10.5. Incompatible materials**

**Incompatible materials** Acids. Bases. Oxidising agent.

**10.6. Hazardous decomposition products**

**Hazardous decomposition products** None known based on information supplied.

## SECTION 11: Toxicological information

**11.1. Information on toxicological effects**

**Information on likely routes of exposure**

**Product Information**

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>   | Specific test data for the substance or mixture is not available Corrosive by inhalation (based on components). Inhalation of corrosive fumes/gases may cause coughing, choking, headache, dizziness, and weakness for several hours. Pulmonary edema may occur with tightness in the chest, shortness of breath, bluish skin, decreased blood pressure, and increased heart rate. Inhaled corrosive substances can lead to a toxic edema of the lungs. Pulmonary edema can be fatal.                                   |
| <b>Eye contact</b>  | Specific test data for the substance or mixture is not available Causes serious eye damage (based on components). Corrosive to the eyes and may cause severe damage including blindness. May cause irreversible damage to eyes.                                                                                                                                                                                                                                                                                         |
| <b>Skin contact</b> | Specific test data for the substance or mixture is not available Corrosive (based on components). Causes burns.                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Ingestion</b>    | Specific test data for the substance or mixture is not available Causes burns (based on components). Ingestion causes burns of the upper digestive and respiratory tracts. May cause severe burning pain in the mouth and stomach with vomiting and diarrhea of dark blood. Blood pressure may decrease. Brownish or yellowish stains may be seen around the mouth. Swelling of the throat may cause shortness of breath and choking. May cause lung damage if swallowed. May be fatal if swallowed and enters airways. |

**Symptoms related to the physical, chemical and toxicological characteristics**

**Symptoms** Redness. Burning. May cause blindness. Coughing and/ or wheezing.

**Acute toxicity**

**Numerical measures of toxicity**

**Component Information**

| Chemical name | Oral LD50            | Dermal LD50 | Inhalation LC50          |
|---------------|----------------------|-------------|--------------------------|
| Sulfuric acid | = 2140 mg/kg ( Rat ) | -           | = 0.375 mg/L ( Rat ) 4 h |



|  |  |  |  |
|--|--|--|--|
|  |  |  |  |
|--|--|--|--|

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                                          |                                                                                                  |
|------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Skin corrosion/irritation</b>         | Classification based on data available for ingredients. Causes severe skin burns and eye damage. |
| <b>Serious eye damage/eye irritation</b> | Classification based on data available for ingredients. Causes serious eye damage. Causes burns. |
| <b>Respiratory or skin sensitisation</b> | No information available.                                                                        |
| <b>Germ cell mutagenicity</b>            | No information available.                                                                        |
| <b>Carcinogenicity</b>                   | No information available.                                                                        |
| <b>Reproductive toxicity</b>             | No information available.                                                                        |
| <b>STOT - single exposure</b>            | No information available.                                                                        |
| <b>STOT - repeated exposure</b>          | No information available.                                                                        |
| <b>Aspiration hazard</b>                 | No information available.                                                                        |
| <b>Other adverse effects</b>             | No information available.                                                                        |

### **SECTION 12: Ecological information**

#### 12.1. Toxicity

**Ecotoxicity** The environmental impact of this product has not been fully investigated.

| Chemical name | Algae/aquatic plants | Fish                                    | Toxicity to microorganisms | Crustacea |
|---------------|----------------------|-----------------------------------------|----------------------------|-----------|
| Sulfuric acid | -                    | LC50: >500mg/L (96h, Brachydanio rerio) | -                          | -         |

#### 12.2. Persistence and degradability

**Persistence and degradability** No information available.

#### 12.3. Bioaccumulative potential

**Bioaccumulation** There is no data for this product.

#### 12.4. Mobility in soil

**Mobility in soil** No information available.

#### 12.5. Results of PBT and vPvB assessment

**PBT and vPvB assessment**

| Chemical name | PBT and vPvB assessment         |
|---------------|---------------------------------|
| Sulfuric acid | The substance is not PBT / vPvB |

**12.6. Endocrine disrupting properties**

No information available.

**SECTION 13: Disposal considerations****13.1. Waste treatment methods**

|                                            |                                                                                                                 |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Waste from residues/unused products</b> | Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. |
| <b>Contaminated packaging</b>              | Do not reuse empty containers.                                                                                  |

**SECTION 14: Transport information****IATA**

|                                          |                                        |
|------------------------------------------|----------------------------------------|
| <b>14.1 UN number or ID number</b>       | UN2796                                 |
| <b>14.2 UN proper shipping name</b>      | Sulphuric acid solution                |
| <b>14.3 Transport hazard class(es)</b>   | 8                                      |
| <b>14.4 Packing group</b>                | II                                     |
| <b>Description</b>                       | UN2796, Sulphuric acid solution, 8, II |
| <b>14.5 Environmental hazards</b>        | Not applicable                         |
| <b>14.6 Special precautions for user</b> |                                        |
| <b>Special Provisions</b>                | None                                   |

**IMDG**

|                                                                     |                                        |
|---------------------------------------------------------------------|----------------------------------------|
| <b>14.1 UN number or ID number</b>                                  | UN2796                                 |
| <b>14.2 UN proper shipping name</b>                                 | SULPHURIC ACID SOLUTION                |
| <b>14.3 Transport hazard class(es)</b>                              | 8                                      |
| <b>14.4 Packing group</b>                                           | II                                     |
| <b>Description</b>                                                  | UN2796, SULPHURIC ACID SOLUTION, 8, II |
| <b>14.5 Environmental hazards</b>                                   | Not applicable                         |
| <b>14.6 Special precautions for user</b>                            |                                        |
| <b>Special Provisions</b>                                           | None                                   |
| <b>EmS-No.</b>                                                      | F-A, S-B                               |
| <b>14.7 Maritime transport in bulk according to IMO instruments</b> | No information available               |

**RID**

|                                          |                                        |
|------------------------------------------|----------------------------------------|
| <b>14.1 UN number or ID number</b>       | UN2796                                 |
| <b>14.2 UN proper shipping name</b>      | SULPHURIC ACID SOLUTION                |
| <b>14.3 Transport hazard class(es)</b>   | 8                                      |
| <b>14.4 Packing group</b>                | II                                     |
| <b>Description</b>                       | UN2796, SULPHURIC ACID SOLUTION, 8, II |
| <b>14.5 Environmental hazards</b>        | Not applicable                         |
| <b>14.6 Special precautions for user</b> |                                        |
| <b>Special Provisions</b>                | None                                   |
| <b>Classification code</b>               | C1                                     |

**ADR**

|                                          |                                      |
|------------------------------------------|--------------------------------------|
| <b>14.1 UN number or ID number</b>       | 2796                                 |
| <b>14.2 UN proper shipping name</b>      | SULPHURIC ACID SOLUTION              |
| <b>14.3 Transport hazard class(es)</b>   | 8                                    |
| <b>14.4 Packing group</b>                | II                                   |
| <b>Description</b>                       | 2796, SULPHURIC ACID SOLUTION, 8, II |
| <b>14.5 Environmental hazards</b>        | Not applicable                       |
| <b>14.6 Special precautions for user</b> |                                      |
| <b>Special Provisions</b>                | None                                 |

Classification code C1  
Tunnel restriction code (E)

## SECTION 15: Regulatory information

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National regulations

##### Authorisations and/or restrictions on use:

This product does not contain substances subject to authorisation (UK REACH - Annex XIV). This product does not contain substances subject to restriction (UK REACH - Annex XVII).

##### Persistent Organic Pollutants

Not applicable

##### Export Notification requirements

Not applicable

##### Named dangerous substances per COMAH (SI 2015/483 as amended)

Not applicable

##### The Ozone-Depleting Substances Regulations 2015

Not applicable

##### The Biocidal Products Regulations 2001 (as amended)

Not applicable

##### The Water Environment (Water Framework Directive) (England and Wales) Regulations 2017 (as amended)

Not applicable

##### Poisons and Explosive Precursors

| Chemical name | Poisons and Explosive Precursors        |
|---------------|-----------------------------------------|
| Sulfuric acid | Explosive precursor, Regulated 15 % w/w |

#### International Inventories

Contact supplier for inventory compliance status

### 15.2. Chemical safety assessment

#### Chemical Safety Report

No information available

## SECTION 16: Other information

### Key or legend to abbreviations and acronyms used in the safety data sheet

#### Full text of any hazard and/or precautionary statements referred to under Sections 2-15

H314 - Causes severe skin burns and eye damage

H318 - Causes serious eye damage

#### Legend

SVHC: Substances of Very High Concern for Authorisation:

#### Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

TWA TWA (time-weighted average)

Ceiling Maximum limit value

+ Sensitisers

STEL

Sk\*

STEL (Short Term Exposure Limit)

Skin designation

**Classification procedure**

|                                                                 |                       |
|-----------------------------------------------------------------|-----------------------|
| Classification according to Regulation (EC) No. 1272/2008 [CLP] | Method Used           |
| Acute oral toxicity                                             | Calculation method    |
| Acute dermal toxicity                                           | Calculation method    |
| Acute inhalation toxicity - gas                                 | Calculation method    |
| Acute inhalation toxicity - vapour                              | Calculation method    |
| Acute inhalation toxicity - dust/mist                           | Calculation method    |
| Skin corrosion/irritation                                       | On basis of test data |
| Serious eye damage/eye irritation                               | Calculation method    |
| Respiratory sensitisation                                       | Calculation method    |
| Skin sensitisation                                              | Calculation method    |
| Mutagenicity                                                    | Calculation method    |
| Carcinogenicity                                                 | Calculation method    |
| Reproductive toxicity                                           | Calculation method    |
| STOT - single exposure                                          | Calculation method    |
| STOT - repeated exposure                                        | Calculation method    |
| Acute aquatic toxicity                                          | Calculation method    |
| Chronic aquatic toxicity                                        | Calculation method    |
| Aspiration hazard                                               | Calculation method    |
| Ozone                                                           | Calculation method    |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 European Chemicals Agency (ECHA) Committee for Risk Assessment (ECHA\_RAC)  
 European Chemicals Agency (ECHA) (ECHA\_API)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 U.S. National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
 Organisation for Economic Co-operation and Development Screening Information Data Set  
 World Health Organization

**Revision date** 09-Aug-2024

**Revision Note** Significant changes throughout SDS. Review all sections

**This SDS complies with the requirements of UK REACH Regulations SI 2019/758 (as amended)**

**Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work**

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**

**UK SDS version information - XGHS**

UL release:  
 GHS Revision 7  
 2024 Q4

Full text of any hazard and/or precautionary H314 - Causes severe skin burns and eye damage H318 - Causes serious eye damage statements referred to under Sections 2-15

| Chemical name | Classification according to GB CLP (SI 2020/1567 as amended) | Specific concentration limit (SCL)                                              |
|---------------|--------------------------------------------------------------|---------------------------------------------------------------------------------|
| Sulfuric acid | Skin Corr. 1A (H314)<br>Eye Dam. 1 (H318)                    | Eye Irrit. 2 :: 5%≤C<15%<br>Skin Corr. 1A :: C≥15%<br>Skin Irrit. 2 :: 5%≤C<15% |



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
UK REACH Regulations (SI 2019/758 as amended)

Revision date 24-Jul-2024

Revision Number 1.4

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### 1.1. Product identifier

**Product Name** R2 - Concentrated washing solution (20X), 70 mL  
**Catalogue Number(s)** 7361A, 7360S, 7360Z  
**Pure substance/mixture** Mixture

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

**Recommended use** In vitro diagnostic Restricted to professional users  
**Uses advised against** No information available

### 1.3. Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories Ltd  
The Junction  
Station Road  
Watford, WD17 1ET  
UK

For further information, please contact

**Technical Service** 00800 00246 723  
Techsupport.UK@bio-rad.com

### 1.4. Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC UK: 44-870-8200418

## SECTION 2: Hazards identification

### 2.1. Classification of the substance or mixture

Not classified Not a hazardous substance or mixture according to the Globally Harmonised System (GHS)

### 2.2. Label elements

Not classified Not a hazardous substance or mixture according to the Globally Harmonised System (GHS)

#### **Hazard statements**

Not classified Not a hazardous substance or mixture according to the Globally Harmonised System (GHS)  
EUH208 Contains 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone. May produce an allergic reaction.

### 2.3. Other hazards

No information available.

### SECTION 3: Composition/information on ingredients

#### 3.1 Substances

Not applicable

#### 3.2 Mixtures

| Chemical name                                                                                  | Weight-% | EC No (EU Index No) | UK REACH registration number | Classification according to GB CLP (SI 2020/1567 as amended)                                                                                                                                          | Specific concentration limit (SCL)                                                                                                             | M-Factor | M-Factor (long-term) |
|------------------------------------------------------------------------------------------------|----------|---------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone<br>55965-84-9 | 0.00123  | (613-167-00-5)      | -                            | Acute Tox. 3 (H301)<br>Acute Tox. 3 (H311)<br>Acute Tox. 3 (H331)<br>Skin Corr. 1B (H314)<br>Eye Dam. 1 (H318)<br>Skin Sens. 1A (H317) (EUH071)<br>Aquatic Acute 1 (H400)<br>Aquatic Chronic 1 (H410) | Eye Irrit. 2 :: 0.06%≤C<0.6%<br>Skin Corr. 1C :: C≥0.6%<br>Skin Irrit. 2 :: 0.06%≤C<0.6%<br>Skin Sens. 1A :: C≥0.0015%<br>Eye Dam. 1 :: C≥0.6% | 100      | 100                  |

**Full text of H- and EUH-phrases: see section 16**

This product does not contain candidate substances of very high concern at a concentration ≥ 0.1% (UK REACH Article 59)

### SECTION 4: First aid measures

#### 4.1. Description of first aid measures

|                     |                                                                                                                   |
|---------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>   | Remove to fresh air.                                                                                              |
| <b>Eye contact</b>  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a doctor. |
| <b>Skin contact</b> | Wash skin with soap and water. In the case of skin irritation or allergic reactions see a doctor.                 |
| <b>Ingestion</b>    | Clean mouth with water and drink afterwards plenty of water.                                                      |

#### 4.2. Most important symptoms and effects, both acute and delayed

|                 |                                                     |
|-----------------|-----------------------------------------------------|
| <b>Symptoms</b> | Prolonged contact may cause redness and irritation. |
|-----------------|-----------------------------------------------------|

#### 4.3. Indication of any immediate medical attention and special treatment needed

|                        |                        |
|------------------------|------------------------|
| <b>Note to doctors</b> | Treat symptomatically. |
|------------------------|------------------------|

### SECTION 5: Firefighting measures

**5.1. Extinguishing media**

**Suitable Extinguishing Media** Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Large Fire** CAUTION: Use of water spray when fighting fire may be inefficient.

**Unsuitable extinguishing media** Do not scatter spilled material with high pressure water streams.

**5.2. Special hazards arising from the substance or mixture**

**Specific hazards arising from the chemical** No information available.

**5.3. Advice for firefighters**

**Special protective equipment and precautions for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

**SECTION 6: Accidental release measures****6.1. Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Ensure adequate ventilation.

**For emergency responders** Use personal protection recommended in Section 8.

**6.2. Environmental precautions**

**Environmental precautions** See Section 12 for additional Ecological Information.

**6.3. Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Take up mechanically, placing in appropriate containers for disposal.

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

**6.4. Reference to other sections**

**Reference to other sections** See section 8 for more information. See section 13 for more information.

**SECTION 7: Handling and storage****7.1. Precautions for safe handling**

**Advice on safe handling** Ensure adequate ventilation.

**General hygiene considerations** Wash hands before breaks and immediately after handling the product.

**7.2. Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Store according to product and label instructions.

**7.3. Specific end use(s)**

**Risk Management Methods (RMM)** The information required is contained in this Safety Data Sheet.



## SECTION 8: Exposure controls/personal protection

### 8.1. Control parameters

|                                                 |                                                                                                                                                             |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Limits</b>                          | This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies. |
| <b>Biological occupational exposure limits</b>  | This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies.            |
| <b>Derived No Effect Level (DNEL)</b>           | No information available.                                                                                                                                   |
| <b>Predicted No Effect Concentration (PNEC)</b> | No information available.                                                                                                                                   |

### 8.2. Exposure controls

|                             |                           |
|-----------------------------|---------------------------|
| <b>Engineering controls</b> | No information available. |
|-----------------------------|---------------------------|

### Personal protective equipment

|                                 |                                                                                                                                                                          |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye/face protection</b>      | Wear safety glasses with side shields (or goggles).                                                                                                                      |
| <b>Hand protection</b>          | Wear suitable gloves.                                                                                                                                                    |
| <b>Skin and body protection</b> | Wear suitable protective clothing.                                                                                                                                       |
| <b>Respiratory protection</b>   | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |

## SECTION 9: Physical and chemical properties

### 9.1. Information on basic physical and chemical properties

|                        |                          |
|------------------------|--------------------------|
| <b>Physical state</b>  | Liquid                   |
| <b>Appearance</b>      | aqueous solution         |
| <b>Colour</b>          | colourless               |
| <b>Odour</b>           | Odourless.               |
| <b>Odour threshold</b> | No information available |

| <u>Property</u>                                | <u>Values</u>     | <u>Remarks • Method</u> |
|------------------------------------------------|-------------------|-------------------------|
| <b>Melting point / freezing point</b>          | No data available | None known              |
| <b>Initial boiling point and boiling range</b> | 100 °C            |                         |
| <b>Flammability</b>                            | No data available | None known              |
| <b>Flammability Limit in Air</b>               |                   | None known              |
| <b>Upper flammability or explosive limits</b>  | No data available |                         |
| <b>Lower flammability or explosive limits</b>  | No data available |                         |
| <b>Flash point</b>                             | No data available | None known              |
| <b>Autoignition temperature</b>                | No data available | None known              |
| <b>Decomposition temperature</b>               |                   | None known              |
| <b>pH</b>                                      | 7.4               |                         |
| <b>pH (as aqueous solution)</b>                | No data available | None known              |
| <b>Kinematic viscosity</b>                     | No data available | None known              |
| <b>Dynamic viscosity</b>                       | No data available | None known              |
| <b>Water solubility</b>                        | Miscible in water |                         |

|                            |                          |            |
|----------------------------|--------------------------|------------|
| Solubility(ies)            | No data available        | None known |
| Partition coefficient      | No data available        | None known |
| Vapour pressure            | No data available        | None known |
| Relative density           | No data available        | None known |
| Bulk density               | No data available        |            |
| Liquid Density             | No data available        |            |
| Relative vapour density    | No data available        | None known |
| Particle characteristics   |                          |            |
| Particle Size              | No information available |            |
| Particle Size Distribution | No information available |            |

**9.2. Other information****SECTION 10: Stability and reactivity****10.1. Reactivity**

Reactivity No information available.

**10.2. Chemical stability**

Stability Stable under normal conditions.

**Explosion data**

Sensitivity to mechanical impact None.

Sensitivity to static discharge None.

**10.3. Possibility of hazardous reactions**

Possibility of hazardous reactions None under normal processing.

**10.4. Conditions to avoid**

Conditions to avoid None known based on information supplied.

**10.5. Incompatible materials**

Incompatible materials None known based on information supplied.

**10.6. Hazardous decomposition products**

Hazardous decomposition products None known based on information supplied.

**SECTION 11: Toxicological information****11.1. Information on toxicological effects****Information on likely routes of exposure****Product Information**

|              |                                                                                                |
|--------------|------------------------------------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available.                              |
| Eye contact  | Specific test data for the substance or mixture is not available.                              |
| Skin contact | Specific test data for the substance or mixture is not available. Causes mild skin irritation. |
| Ingestion    | Specific test data for the substance or mixture is not available.                              |

**Symptoms related to the physical, chemical and toxicological characteristics**

Symptoms Prolonged contact may cause redness and irritation.

**Acute toxicity****Numerical measures of toxicity**

The following values are calculated based on chapter 3.1 of the GHS document

ATEmix (oral) 215,413.50 mg/kg  
ATEmix (dermal) 206,611.60 mg/kg

**Component Information**

| Chemical name                                                                    | Oral LD50          | Dermal LD50              | Inhalation LC50 |
|----------------------------------------------------------------------------------|--------------------|--------------------------|-----------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone | = 53 mg/kg ( Rat ) | = 87.12 mg/kg ( Rabbit ) | -               |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Skin corrosion/irritation** Classification based on data available for ingredients. Causes mild skin irritation.

**Serious eye damage/eye irritation** No information available.

**Respiratory or skin sensitisation** No information available.

**Germ cell mutagenicity** No information available.

**Carcinogenicity** No information available.

**Reproductive toxicity** No information available.

**STOT - single exposure** No information available.

**STOT - repeated exposure** No information available.

**Aspiration hazard** No information available.

**Other adverse effects** No information available.

## SECTION 12: Ecological information

**12.1. Toxicity**

**Ecotoxicity** The environmental impact of this product has not been fully investigated.

**12.2. Persistence and degradability**

**Persistence and degradability** No information available.

**12.3. Bioaccumulative potential****Bioaccumulation****Component Information**

| Chemical name | Partition coefficient |
|---------------|-----------------------|
|---------------|-----------------------|

|                                                                                  |     |
|----------------------------------------------------------------------------------|-----|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone | 0.7 |
|----------------------------------------------------------------------------------|-----|

**12.4. Mobility in soil**

Mobility in soil No information available.

**12.5. Results of PBT and vPvB assessment**

PBT and vPvB assessment No information available.

| Chemical name                                                                    | PBT and vPvB assessment         |
|----------------------------------------------------------------------------------|---------------------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone | The substance is not PBT / vPvB |

**12.6. Endocrine disrupting properties**

No information available.

## SECTION 13: Disposal considerations

**13.1. Waste treatment methods**

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

## SECTION 14: Transport information

**IATA**

14.1 UN number or ID number Not regulated  
 14.2 UN proper shipping name Not regulated  
 14.3 Transport hazard class(es) Not regulated  
 14.4 Packing group Not regulated  
 14.5 Environmental hazards Not applicable  
 14.6 Special precautions for user  
 Special Provisions None

**IMDG**

14.1 UN number or ID number Not regulated  
 14.2 UN proper shipping name Not regulated  
 14.3 Transport hazard class(es) Not regulated  
 14.4 Packing group Not regulated  
 14.5 Environmental hazards Not applicable  
 14.6 Special precautions for user  
 Special Provisions None  
 14.7 Maritime transport in bulk according to IMO instruments No information available

**RID**

14.1 UN number or ID number Not regulated  
 14.2 UN proper shipping name Not regulated  
 14.3 Transport hazard class(es) Not regulated  
 14.4 Packing group Not regulated  
 14.5 Environmental hazards Not applicable  
 14.6 Special precautions for user  
 Special Provisions None

**ADR**

14.1 UN number or ID number Not regulated  
 14.2 UN proper shipping name Not regulated

|                                          |                |
|------------------------------------------|----------------|
| <b>14.3 Transport hazard class(es)</b>   | Not regulated  |
| <b>14.4 Packing group</b>                | Not regulated  |
| <b>14.5 Environmental hazards</b>        | Not applicable |
| <b>14.6 Special precautions for user</b> |                |
| <b>Special Provisions</b>                | None           |

## SECTION 15: Regulatory information

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National regulations

##### **Authorisations and/or restrictions on use:**

This product does not contain substances subject to authorisation (UK REACH - Annex XIV). This product does not contain substances subject to restriction (UK REACH - Annex XVII).

##### **Persistent Organic Pollutants**

Not applicable

##### **Export Notification requirements**

Not applicable

##### **Named dangerous substances per COMAH (SI 2015/483 as amended)**

Not applicable

##### **The Ozone-Depleting Substances Regulations 2015**

Not applicable

##### **The Biocidal Products Regulations 2001 (as amended)**

| Chemical name                                                                                    | The Biocidal Products Regulations 2001 (as amended) |
|--------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone - 55965-84-9 | PT2<br>PT4<br>PT6<br>PT11<br>PT12<br>PT13           |

##### **The Water Environment (Water Framework Directive) (England and Wales) Regulations 2017 (as amended)**

Not applicable

##### **Poisons and Explosive Precursors**

Not applicable

#### International Inventories

Contact supplier for inventory compliance status

### 15.2. Chemical safety assessment

#### **Chemical Safety Report**

No information available

## SECTION 16: Other information

### Key or legend to abbreviations and acronyms used in the safety data sheet

#### **Full text of any hazard and/or precautionary statements referred to under Sections 2-15**

EUH071 - Corrosive to the respiratory tract

H301 - Toxic if swallowed

H311 - Toxic in contact with skin

H314 - Causes severe skin burns and eye damage

H317 - May cause an allergic skin reaction

H318 - Causes serious eye damage  
H331 - Toxic if inhaled  
H400 - Very toxic to aquatic life  
H410 - Very toxic to aquatic life with long lasting effects

**Legend**

SVHC: Substances of Very High Concern for Authorisation:

**Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | Sk*  | Skin designation                 |
| +       | Sensitisers                 |      |                                  |

**Classification procedure**

|                                                                 |                    |
|-----------------------------------------------------------------|--------------------|
| Classification according to Regulation (EC) No. 1272/2008 [CLP] | Method Used        |
| Acute oral toxicity                                             | Calculation method |
| Acute dermal toxicity                                           | Calculation method |
| Acute inhalation toxicity - gas                                 | Calculation method |
| Acute inhalation toxicity - vapour                              | Calculation method |
| Acute inhalation toxicity - dust/mist                           | Calculation method |
| Skin corrosion/irritation                                       | Calculation method |
| Serious eye damage/eye irritation                               | Calculation method |
| Respiratory sensitisation                                       | Calculation method |
| Skin sensitisation                                              | Calculation method |
| Mutagenicity                                                    | Calculation method |
| Carcinogenicity                                                 | Calculation method |
| Reproductive toxicity                                           | Calculation method |
| STOT - single exposure                                          | Calculation method |
| STOT - repeated exposure                                        | Calculation method |
| Acute aquatic toxicity                                          | Calculation method |
| Chronic aquatic toxicity                                        | Calculation method |
| Aspiration hazard                                               | Calculation method |
| Ozone                                                           | Calculation method |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
European Chemicals Agency (ECHA) Committee for Risk Assessment (ECHA\_RAC)  
European Chemicals Agency (ECHA) (ECHA\_API)  
Environmental Protection Agency  
Acute Exposure Guideline Level(s) (AEGL(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
National Institute of Technology and Evaluation (NITE)  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
U.S. National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
World Health Organization

Revision date 24-Jul-2024

**Revision Note**

Significant changes throughout SDS. Review all sections

**This SDS complies with the requirements of UK REACH Regulations SI 2019/758 (as amended)**

**Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work**

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

End of Safety Data Sheet

UK SDS version information - XGHS

UL release:  
GHS Revision 7  
2024 Q4

Full text of any hazard and/or precautionary EUH071 - Corrosive to the respiratory tract H301 - Toxic if swallowed H311 - Toxic in contact with skin  
statements referred to under Sections 2-15 H314 - Causes severe skin burns and eye damage H317 - May cause an allergic skin reaction H318 - Causes serious eye damage H331 - Toxic if inhaled H400 - Very toxic to aquatic life H410 - Very toxic to aquatic life with long lasting effects

| Chemical name                                                                    | Classification according to GB CLP (SI 2020/1567 as amended)                                                                                                                                             | Specific concentration limit (SCL)                                                                                                                  |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone | Acute Tox. 3 (H301)<br>Acute Tox. 3 (H311)<br>Acute Tox. 3 (H331)<br>Skin Corr. 1B (H314)<br>Eye Dam. 1 (H318)<br>Skin Sens. 1A (H317)<br>(EUH071)<br>Aquatic Acute 1 (H400)<br>Aquatic Chronic 1 (H410) | Eye Irrit. 2 :: 0.06%<=C<0.6%<br>Skin Corr. 1C :: C>=0.6%<br>Skin Irrit. 2 :: 0.06%<=C<0.6%<br>Skin Sens. 1A :: C>=0.0015%<br>Eye Dam. 1 :: C>=0.6% |



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
UK REACH Regulations (SI 2019/758 as amended)

Revision date 06-Aug-2024

Revision Number 1.2

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### 1.1. Product identifier

**Product Name** R9 - Chromogen: TMB Solution (11X)  
**Catalogue Number(s)** 7436L, 7436H, 26182, 7337T  
**Pure substance/mixture** Mixture

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

**Recommended use** Restricted to professional users In vitro diagnostic  
**Uses advised against** No information available

### 1.3. Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories Ltd  
The Junction  
Station Road  
Watford, WD17 1ET  
UK

For further information, please contact

**Technical Service** 00800 00246 723  
Techsupport.UK@bio-rad.com

### 1.4. Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC UK: 44-870-8200418

## SECTION 2: Hazards identification

### 2.1. Classification of the substance or mixture

|                                  |                     |
|----------------------------------|---------------------|
| <b>Skin corrosion/irritation</b> | Category 1 - (H314) |
|----------------------------------|---------------------|

### 2.2. Label elements



**Signal word**  
Danger

#### **Hazard statements**

H314 - Causes severe skin burns and eye damage



**Precautionary statements**

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
 P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
 P301 + P330 + P331 - IF SWALLOWED: Rinse mouth. Do NOT induce vomiting  
 P501 - Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable  
 P280 - Wear protective gloves, protective clothing, eye protection and face protection

**2.3. Other hazards**

No information available.

**SECTION 3: Composition/information on ingredients****3.1 Substances**

Not applicable

**3.2 Mixtures**

| Chemical name                  | Weight-% | EC No (EU Index No)         | UK REACH registration number | Classification according to GB CLP (SI 2020/1567 as amended)    | Specific concentration limit (SCL)                                                                 | M-Factor | M-Factor (long-term) |
|--------------------------------|----------|-----------------------------|------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------|----------|----------------------|
| Hydrochloric acid<br>7647-01-0 | 0.3      | 231-595-7<br>(017-002-00-2) | -                            | Skin Corr. 1B (H314)<br>Eye Irrit. 2 (H319)<br>STOT SE 3 (H335) | Eye Irrit. 2 :: 1%≤C<3%<br>Skin Corr. 1B :: C≥5%<br>Skin Irrit. 2 :: 1%≤C<5%<br>STOT SE 3 :: C≥10% | -        | -                    |
| Ethyl alcohol<br>64-17-5       | 0.077    | 200-578-6<br>(603-002-00-5) | -                            | Flam. Liq. 2 (H225)                                             | -                                                                                                  | -        | -                    |

**Full text of H- and EUH-phrases: see section 16**

This product does not contain candidate substances of very high concern at a concentration ≥ 0.1% (UK REACH Article 59)

**SECTION 4: First aid measures****4.1. Description of first aid measures**

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b> | Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.                                                                                                                                                                                                                                                                                                                                      |
| <b>Inhalation</b>     | Remove to fresh air. If breathing has stopped, give artificial respiration. Get medical attention immediately. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. If breathing is difficult, (trained personnel should) give oxygen. Delayed pulmonary edema may occur. |
| <b>Eye contact</b>    | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Keep eye wide open while rinsing. Do not rub affected area. Remove contact lenses, if present and easy to do. Continue rinsing. Get immediate medical attention.                                                                                                                                                                              |
| <b>Skin contact</b>   | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Get immediate medical attention.                                                                                                                                                                                                                                                                                                 |

|                                    |                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ingestion                          | Do NOT induce vomiting. Rinse mouth. Never give anything by mouth to an unconscious person. Get immediate medical attention.                                                                                                                                                                                                     |
| Self-protection of the first aider | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. Avoid contact with skin, eyes or clothing. Avoid direct contact with skin. Use barrier to give mouth-to-mouth resuscitation. Wear personal protective clothing (see section 8). |

#### **4.2. Most important symptoms and effects, both acute and delayed**

|          |                    |
|----------|--------------------|
| Symptoms | Burning sensation. |
|----------|--------------------|

#### **4.3. Indication of any immediate medical attention and special treatment needed**

|                 |                                                                                                                                                                                                                                                                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Note to doctors | Product is a corrosive material. Use of gastric lavage or emesis is contra-indicated. Possible perforation of stomach or esophagus should be investigated. Do not give chemical antidotes. Asphyxia from glottal edema may occur. Marked decrease in blood pressure may occur with moist rales, frothy sputum, and high pulse pressure. |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### **SECTION 5: Firefighting measures**

#### **5.1. Extinguishing media**

|                              |                                                                                                         |
|------------------------------|---------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
|------------------------------|---------------------------------------------------------------------------------------------------------|

|            |                                                                    |
|------------|--------------------------------------------------------------------|
| Large Fire | CAUTION: Use of water spray when fighting fire may be inefficient. |
|------------|--------------------------------------------------------------------|

|                                |                                                                   |
|--------------------------------|-------------------------------------------------------------------|
| Unsuitable extinguishing media | Do not scatter spilled material with high pressure water streams. |
|--------------------------------|-------------------------------------------------------------------|

#### **5.2. Special hazards arising from the substance or mixture**

|                                            |                                                                                                                                         |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Specific hazards arising from the chemical | The product causes burns of eyes, skin and mucous membranes. Thermal decomposition can lead to release of irritating gases and vapours. |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|

#### **5.3. Advice for firefighters**

|                                                                |                                                                                                                                    |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|

### **SECTION 6: Accidental release measures**

#### **6.1. Personal precautions, protective equipment and emergency procedures**

|                      |                                                                                                                                                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Personal precautions | Attention! Corrosive material. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                   |                                                          |
|-------------------|----------------------------------------------------------|
| Other information | Refer to protective measures listed in Sections 7 and 8. |
|-------------------|----------------------------------------------------------|

|                          |                                                   |
|--------------------------|---------------------------------------------------|
| For emergency responders | Use personal protection recommended in Section 8. |
|--------------------------|---------------------------------------------------|

#### **6.2. Environmental precautions**

|                           |                                                                                                                                                                                   |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Environmental precautions | Prevent further leakage or spillage if safe to do so. Should not be released into the environment. Do not allow to enter into soil/subsoil. Prevent product from entering drains. |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### **6.3. Methods and material for containment and cleaning up**

|                         |                                                       |
|-------------------------|-------------------------------------------------------|
| Methods for containment | Prevent further leakage or spillage if safe to do so. |
|-------------------------|-------------------------------------------------------|

|                         |                                                                       |
|-------------------------|-----------------------------------------------------------------------|
| Methods for cleaning up | Take up mechanically, placing in appropriate containers for disposal. |
|-------------------------|-----------------------------------------------------------------------|

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

#### 6.4. Reference to other sections

**Reference to other sections** See section 8 for more information. See section 13 for more information.

## SECTION 7: Handling and storage

### 7.1. Precautions for safe handling

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. In case of insufficient ventilation, wear suitable respiratory equipment. Handle product only in closed system or provide appropriate exhaust ventilation. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash it before reuse.

**General hygiene considerations** Avoid contact with skin, eyes or clothing. Wear suitable gloves and eye/face protection. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Contaminated work clothing should not be allowed out of the workplace. Regular cleaning of equipment, work area and clothing is recommended. Wash hands before breaks and immediately after handling the product.

### 7.2. Conditions for safe storage, including any incompatibilities

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place. Protect from moisture. Store locked up. Keep out of the reach of children. Store away from other materials. Store according to product and label instructions.

### 7.3. Specific end use(s)

**Risk Management Methods (RMM)** The information required is contained in this Safety Data Sheet.

## SECTION 8: Exposure controls/personal protection

### 8.1. Control parameters

#### Exposure Limits

| Chemical name                  | United Kingdom                                                                                 |
|--------------------------------|------------------------------------------------------------------------------------------------|
| Hydrochloric acid<br>7647-01-0 | TWA: 1 ppm<br>TWA: 2 mg/m <sup>3</sup><br>STEL: 5 ppm<br>STEL: 8 mg/m <sup>3</sup>             |
| Ethyl alcohol<br>64-17-5       | TWA: 1000 ppm<br>TWA: 1920 mg/m <sup>3</sup><br>STEL: 3000 ppm<br>STEL: 5760 mg/m <sup>3</sup> |

**Biological occupational exposure limits** This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies.

**Derived No Effect Level (DNEL)** No information available.

**Predicted No Effect Concentration (PNEC)** No information available.

### 8.2. Exposure controls

|                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Engineering controls</b>                 | No information available.                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b><u>Personal protective equipment</u></b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Eye/face protection</b>                  | Tight sealing safety goggles. Face protection shield.                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Hand protection</b>                      | Wear suitable gloves. Impervious gloves.                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Skin and body protection</b>             | Wear suitable protective clothing. Long sleeved clothing. Chemical resistant apron.                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Respiratory protection</b>               | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.                                                                                                                                                                                                                                                                              |
| <b>General hygiene considerations</b>       | Avoid contact with skin, eyes or clothing. Wear suitable gloves and eye/face protection. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Contaminated work clothing should not be allowed out of the workplace. Regular cleaning of equipment, work area and clothing is recommended. Wash hands before breaks and immediately after handling the product. |

## SECTION 9: Physical and chemical properties

### 9.1. Information on basic physical and chemical properties

|                        |                          |
|------------------------|--------------------------|
| <b>Physical state</b>  | Liquid                   |
| <b>Appearance</b>      | Liquid                   |
| <b>Colour</b>          | pink                     |
| <b>Odour</b>           | Low.                     |
| <b>Odour threshold</b> | No information available |

| <b><u>Property</u></b>                         | <b><u>Values</u></b>     | <b><u>Remarks • Method</u></b> |
|------------------------------------------------|--------------------------|--------------------------------|
| <b>Melting point / freezing point</b>          | No data available        | None known                     |
| <b>Initial boiling point and boiling range</b> | No data available        | None known                     |
| <b>Flammability</b>                            | No data available        | None known                     |
| <b>Flammability Limit in Air</b>               |                          | None known                     |
| <b>Upper flammability or explosive limits</b>  | No data available        |                                |
| <b>Lower flammability or explosive limits</b>  | No data available        |                                |
| <b>Flash point</b>                             | No data available        | None known                     |
| <b>Autoignition temperature</b>                | No data available        |                                |
| <b>Decomposition temperature</b>               |                          | None known                     |
| <b>pH</b>                                      | 1.5                      |                                |
| <b>pH (as aqueous solution)</b>                | No data available        | None known                     |
| <b>Kinematic viscosity</b>                     | No data available        | None known                     |
| <b>Dynamic viscosity</b>                       | No data available        | None known                     |
| <b>Water solubility</b>                        | Miscible in water        |                                |
| <b>Solubility(ies)</b>                         | No data available        | None known                     |
| <b>Partition coefficient</b>                   | No data available        | None known                     |
| <b>Vapour pressure</b>                         | No data available        | None known                     |
| <b>Relative density</b>                        | No data available        | None known                     |
| <b>Bulk density</b>                            | No data available        |                                |
| <b>Liquid Density</b>                          | No data available        |                                |
| <b>Relative vapour density</b>                 | No data available        | None known                     |
| <b>Particle characteristics</b>                |                          |                                |
| <b>Particle Size</b>                           | No information available |                                |
| <b>Particle Size Distribution</b>              | No information available |                                |

### 9.2. Other information

## SECTION 10: Stability and reactivity

**10.1. Reactivity**

Reactivity No information available.

**10.2. Chemical stability**

Stability Stable under normal conditions.

**Explosion data**

Sensitivity to mechanical impact None.

Sensitivity to static discharge None.

**10.3. Possibility of hazardous reactions**

Possibility of hazardous reactions None under normal processing.

**10.4. Conditions to avoid**

Conditions to avoid Exposure to air or moisture over prolonged periods.

**10.5. Incompatible materials**

Incompatible materials Acids. Bases. Oxidising agent.

**10.6. Hazardous decomposition products**

Hazardous decomposition products None known based on information supplied.

**SECTION 11: Toxicological information****11.1. Information on toxicological effects****Information on likely routes of exposure****Product Information**

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available Corrosive by inhalation (based on components). Inhalation of corrosive fumes/gases may cause coughing, choking, headache, dizziness, and weakness for several hours. Pulmonary edema may occur with tightness in the chest, shortness of breath, bluish skin, decreased blood pressure, and increased heart rate. Inhaled corrosive substances can lead to a toxic edema of the lungs. Pulmonary edema can be fatal.                                   |
| Eye contact  | Specific test data for the substance or mixture is not available Causes serious eye damage (based on components). Corrosive to the eyes and may cause severe damage including blindness.                                                                                                                                                                                                                                                                                                                                |
| Skin contact | Specific test data for the substance or mixture is not available Corrosive (based on components). Causes burns.                                                                                                                                                                                                                                                                                                                                                                                                         |
| Ingestion    | Specific test data for the substance or mixture is not available Causes burns (based on components). Ingestion causes burns of the upper digestive and respiratory tracts. May cause severe burning pain in the mouth and stomach with vomiting and diarrhea of dark blood. Blood pressure may decrease. Brownish or yellowish stains may be seen around the mouth. Swelling of the throat may cause shortness of breath and choking. May cause lung damage if swallowed. May be fatal if swallowed and enters airways. |

**Symptoms related to the physical, chemical and toxicological characteristics**

Symptoms Redness. Burning. May cause blindness. Coughing and/ or wheezing.

**Acute toxicity****Numerical measures of toxicity**

**Component Information**

| Chemical name     | Oral LD50               | Dermal LD50             | Inhalation LC50                                      |
|-------------------|-------------------------|-------------------------|------------------------------------------------------|
| Hydrochloric acid | 238 - 277 mg/kg ( Rat ) | > 5010 mg/kg ( Rabbit ) | = 1.68 mg/L ( Rat ) 1 h                              |
| Ethyl alcohol     | = 7060 mg/kg ( Rat )    | -                       | = 116.9 mg/L ( Rat ) 4 h<br>= 133.8 mg/L ( Rat ) 4 h |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                                          |                                                                                                  |
|------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Skin corrosion/irritation</b>         | Classification based on data available for ingredients. Causes severe skin burns and eye damage. |
| <b>Serious eye damage/eye irritation</b> | Classification based on data available for ingredients. Causes serious eye damage.               |
| <b>Respiratory or skin sensitisation</b> | No information available.                                                                        |
| <b>Germ cell mutagenicity</b>            | No information available.                                                                        |
| <b>Carcinogenicity</b>                   | No information available.                                                                        |
| <b>Reproductive toxicity</b>             | No information available.                                                                        |
| <b>STOT - single exposure</b>            | No information available.                                                                        |
| <b>STOT - repeated exposure</b>          | No information available.                                                                        |
| <b>Aspiration hazard</b>                 | No information available.                                                                        |
| <b>Other adverse effects</b>             | No information available.                                                                        |

**SECTION 12: Ecological information****12.1. Toxicity**

**Ecotoxicity** The environmental impact of this product has not been fully investigated.

| Chemical name | Algae/aquatic plants | Fish                                                                                                                                                | Toxicity to microorganisms | Crustacea                                                                        |
|---------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------------------------------------------------------------------------|
| Ethyl alcohol | -                    | LC50: 12.0 - 16.0mL/L (96h, Oncorhynchus mykiss)<br>LC50: >100mg/L (96h, Pimephales promelas)<br>LC50: 13400 - 15100mg/L (96h, Pimephales promelas) | -                          | LC50: 9268 - 14221mg/L (48h, Daphnia magna)<br>EC50: =2mg/L (48h, Daphnia magna) |

**12.2. Persistence and degradability**

**Persistence and degradability** No information available.

**12.3. Bioaccumulative potential****Bioaccumulation****Component Information**

| Chemical name | Partition coefficient |
|---------------|-----------------------|
| Ethyl alcohol | -0.35                 |

**12.4. Mobility in soil**

**Mobility in soil** No information available.

**12.5. Results of PBT and vPvB assessment**

**PBT and vPvB assessment** No information available.

| Chemical name     | PBT and vPvB assessment         |
|-------------------|---------------------------------|
| Hydrochloric acid | The substance is not PBT / vPvB |
| Ethyl alcohol     | The substance is not PBT / vPvB |

**12.6. Endocrine disrupting properties**

No information available.

## SECTION 13: Disposal considerations

**13.1. Waste treatment methods**

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

## SECTION 14: Transport information

**IATA**

**14.1 UN number or ID number** Not regulated

**14.2 UN proper shipping name** Not regulated

**14.3 Transport hazard class(es)** Not regulated

**14.4 Packing group** Not regulated

**14.5 Environmental hazards** Not applicable

**14.6 Special precautions for user**

**Special Provisions** None

**IMDG**

**14.1 UN number or ID number** Not regulated

**14.2 UN proper shipping name** Not regulated

**14.3 Transport hazard class(es)** Not regulated

**14.4 Packing group** Not regulated

**14.5 Environmental hazards** Not applicable

**14.6 Special precautions for user**

**Special Provisions** None

**14.7 Maritime transport in bulk according to IMO instruments** No information available

**RID**

**14.1 UN number or ID number** Not regulated

**14.2 UN proper shipping name** Not regulated

**14.3 Transport hazard class(es)** Not regulated

**14.4 Packing group** Not regulated

**14.5 Environmental hazards** Not applicable

**14.6 Special precautions for user**

Special Provisions None

**ADR**

14.1 UN number or ID number Not regulated  
 14.2 UN proper shipping name Not regulated  
 14.3 Transport hazard class(es) Not regulated  
 14.4 Packing group Not regulated  
 14.5 Environmental hazards Not applicable  
 14.6 Special precautions for user  
 Special Provisions None

## SECTION 15: Regulatory information

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National regulations

##### Authorisations and/or restrictions on use:

This product does not contain substances subject to authorisation (UK REACH - Annex XIV). This product does not contain substances subject to restriction (UK REACH - Annex XVII).

##### Persistent Organic Pollutants

Not applicable

##### Export Notification requirements

Not applicable

##### Named dangerous substances per COMAH (SI 2015/483 as amended)

| Chemical name                 | Lower-tier requirements (tons) | Upper-tier requirements (tons) |
|-------------------------------|--------------------------------|--------------------------------|
| Hydrochloric acid - 7647-01-0 | 25                             | 250                            |

##### The Ozone-Depleting Substances Regulations 2015

Not applicable

##### The Biocidal Products Regulations 2001 (as amended)

| Chemical name                 | The Biocidal Products Regulations 2001 (as amended) |
|-------------------------------|-----------------------------------------------------|
| Hydrochloric acid - 7647-01-0 | PT2                                                 |
| Ethyl alcohol - 64-17-5       | PT2<br>PT4<br>PT1                                   |

##### The Water Environment (Water Framework Directive) (England and Wales) Regulations 2017 (as amended)

Not applicable

##### Poisons and Explosive Precursors

| Chemical name     | Poisons and Explosive Precursors        |
|-------------------|-----------------------------------------|
| Hydrochloric acid | Explosive precursor, Regulated 10 % w/w |

#### International Inventories

Contact supplier for inventory compliance status

### 15.2. Chemical safety assessment

#### Chemical Safety Report

No information available

## SECTION 16: Other information



**Key or legend to abbreviations and acronyms used in the safety data sheet****Full text of any hazard and/or precautionary statements referred to under Sections 2-15**

H225 - Highly flammable liquid and vapour

H314 - Causes severe skin burns and eye damage

H319 - Causes serious eye irritation

H335 - May cause respiratory irritation

**Legend**

SVHC: Substances of Very High Concern for Authorisation:

**Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | Sk*  | Skin designation                 |
| +       | Sensitisers                 |      |                                  |

**Classification procedure**

| Classification according to Regulation (EC) No. 1272/2008 [CLP] | Method Used           |
|-----------------------------------------------------------------|-----------------------|
| Acute oral toxicity                                             | Calculation method    |
| Acute dermal toxicity                                           | Calculation method    |
| Acute inhalation toxicity - gas                                 | Calculation method    |
| Acute inhalation toxicity - vapour                              | Calculation method    |
| Acute inhalation toxicity - dust/mist                           | Calculation method    |
| Skin corrosion/irritation                                       | On basis of test data |
| Serious eye damage/eye irritation                               | Calculation method    |
| Respiratory sensitisation                                       | Calculation method    |
| Skin sensitisation                                              | Calculation method    |
| Mutagenicity                                                    | Calculation method    |
| Carcinogenicity                                                 | Calculation method    |
| Reproductive toxicity                                           | Calculation method    |
| STOT - single exposure                                          | Calculation method    |
| STOT - repeated exposure                                        | Calculation method    |
| Acute aquatic toxicity                                          | Calculation method    |
| Chronic aquatic toxicity                                        | Calculation method    |
| Aspiration hazard                                               | Calculation method    |
| Ozone                                                           | Calculation method    |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 European Chemicals Agency (ECHA) Committee for Risk Assessment (ECHA\_RAC)  
 European Chemicals Agency (ECHA) (ECHA\_API)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 U.S. National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
 Organisation for Economic Co-operation and Development Screening Information Data Set  
 World Health Organization

Revision date 06-Aug-2024

Revision Note Significant changes throughout SDS. Review all sections

This SDS complies with the requirements of UK REACH Regulations SI 2019/758 (as amended)

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**

**UK SDS version information - XGHS**

UL release:

GHS Revision 7

2024 Q4

Full text of any hazard and/or precautionary H225 - Highly flammable liquid and vapour H314 - Causes severe skin burns and eye damage H319 - statements referred to under Sections 2-15 Causes serious eye irritation H335 - May cause respiratory irritation

| Chemical name     | Classification according to GB CLP (SI 2020/1567 as amended)    | Specific concentration limit (SCL)                                                                 |
|-------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Hydrochloric acid | Skin Corr. 1B (H314)<br>Eye Irrit. 2 (H319)<br>STOT SE 3 (H335) | Eye Irrit. 2 :: 1%≤C<3%<br>Skin Corr. 1B :: C≥5%<br>Skin Irrit. 2 :: 1%≤C<5%<br>STOT SE 3 :: C≥10% |
| Ethyl alcohol     | Flam. Liq. 2 (H225)                                             |                                                                                                    |



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
UK REACH Regulations (SI 2019/758 as amended)

Revision date 12-Dec-2024

Revision Number 1.1

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### 1.1. Product identifier

**Product Name** R1 - Microplate, 12 strips x 8 wells  
**Catalogue Number(s)** 7253A  
**Pure substance/mixture** Mixture

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

**Recommended use** Restricted to professional users In vitro diagnostic  
**Uses advised against** No information available

### 1.3. Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories Ltd  
The Junction  
Station Road  
Watford, WD17 1ET  
UK

For further information, please contact

**Technical Service** 00800 00246 723  
Techsupport.UK@bio-rad.com

### 1.4. Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC UK: 44-870-8200418

## SECTION 2: Hazards identification

### 2.1. Classification of the substance or mixture

Not classified Not a hazardous substance or mixture according to the Globally Harmonised System (GHS)

### 2.2. Label elements

Not classified Not a hazardous substance or mixture according to the Globally Harmonised System (GHS)

#### **Hazard statements**

Not classified Not a hazardous substance or mixture according to the Globally Harmonised System (GHS)

### 2.3. Other hazards

No information available.

## SECTION 3: Composition/information on ingredients

### **3.1 Substances**

The product contains no substances which at their given concentration, are considered to be hazardous to health

### **Full text of H- and EUH-phrases: see section 16**

This product does not contain candidate substances of very high concern at a concentration  $\geq 0.1\%$  (UK REACH Article 59)

## **SECTION 4: First aid measures**

### **4.1. Description of first aid measures**

|              |                                                                                                                   |
|--------------|-------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                              |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a doctor. |
| Skin contact | Wash skin with soap and water. In the case of skin irritation or allergic reactions see a doctor.                 |
| Ingestion    | Rinse mouth.                                                                                                      |

### **4.2. Most important symptoms and effects, both acute and delayed**

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

### **4.3. Indication of any immediate medical attention and special treatment needed**

|                 |                        |
|-----------------|------------------------|
| Note to doctors | Treat symptomatically. |
|-----------------|------------------------|

## **SECTION 5: Firefighting measures**

### **5.1. Extinguishing media**

|                              |                                                                                                         |
|------------------------------|---------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
|------------------------------|---------------------------------------------------------------------------------------------------------|

|            |                                                                    |
|------------|--------------------------------------------------------------------|
| Large Fire | CAUTION: Use of water spray when fighting fire may be inefficient. |
|------------|--------------------------------------------------------------------|

|                                |                                                                   |
|--------------------------------|-------------------------------------------------------------------|
| Unsuitable extinguishing media | Do not scatter spilled material with high pressure water streams. |
|--------------------------------|-------------------------------------------------------------------|

### **5.2. Special hazards arising from the substance or mixture**

|                                            |                           |
|--------------------------------------------|---------------------------|
| Specific hazards arising from the chemical | No information available. |
|--------------------------------------------|---------------------------|

### **5.3. Advice for firefighters**

|                                                                |                                                                                                                                    |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|

## **SECTION 6: Accidental release measures**

### **6.1. Personal precautions, protective equipment and emergency procedures**

|                      |                              |
|----------------------|------------------------------|
| Personal precautions | Ensure adequate ventilation. |
|----------------------|------------------------------|

**For emergency responders** Use personal protection recommended in Section 8.

#### **6.2. Environmental precautions**

**Environmental precautions** See Section 12 for additional Ecological Information.

#### **6.3. Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Take up mechanically, placing in appropriate containers for disposal.

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

#### **6.4. Reference to other sections**

**Reference to other sections** See section 8 for more information. See section 13 for more information.

### **SECTION 7: Handling and storage**

#### **7.1. Precautions for safe handling**

**Advice on safe handling** Ensure adequate ventilation.

**General hygiene considerations** Wash hands before breaks and immediately after handling the product.

#### **7.2. Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Store according to product and label instructions.

#### **7.3. Specific end use(s)**

**Risk Management Methods (RMM)** The information required is contained in this Safety Data Sheet.

### **SECTION 8: Exposure controls/personal protection**

#### **8.1. Control parameters**

**Exposure Limits** This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Biological occupational exposure limits** This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies.

**Derived No Effect Level (DNEL)** No information available.

**Predicted No Effect Concentration (PNEC)** No information available.

#### **8.2. Exposure controls**

**Engineering controls** No information available.

#### **Personal protective equipment**

**Eye/face protection** No special protective equipment required.

|                                 |                                                                                                                                                                          |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b> | No special protective equipment required.                                                                                                                                |
| <b>Respiratory protection</b>   | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |

## SECTION 9: Physical and chemical properties

### 9.1. Information on basic physical and chemical properties

|                        |                          |
|------------------------|--------------------------|
| <b>Physical state</b>  | Solid                    |
| <b>Appearance</b>      | solid                    |
| <b>Colour</b>          | colourless               |
| <b>Odour</b>           | Odourless.               |
| <b>Odour threshold</b> | No information available |

| <u>Property</u>                                | <u>Values</u>            | <u>Remarks • Method</u> |
|------------------------------------------------|--------------------------|-------------------------|
| <b>Melting point / freezing point</b>          | No data available        | None known              |
| <b>Initial boiling point and boiling range</b> | No data available        | None known              |
| <b>Flammability</b>                            | No data available        | None known              |
| <b>Flammability Limit in Air</b>               |                          | None known              |
| <b>Upper flammability or explosive limits</b>  | No data available        |                         |
| <b>Lower flammability or explosive limits</b>  | No data available        |                         |
| <b>Flash point</b>                             | No data available        | None known              |
| <b>Autoignition temperature</b>                | No data available        | None known              |
| <b>Decomposition temperature</b>               |                          | None known              |
| <b>pH</b>                                      |                          | None known              |
| <b>pH (as aqueous solution)</b>                | No data available        | None known              |
| <b>Kinematic viscosity</b>                     | No data available        | None known              |
| <b>Dynamic viscosity</b>                       | No data available        | None known              |
| <b>Water solubility</b>                        | Insoluble in water       |                         |
| <b>Solubility(ies)</b>                         | No data available        | None known              |
| <b>Partition coefficient</b>                   | No data available        | None known              |
| <b>Vapour pressure</b>                         | No data available        | None known              |
| <b>Relative density</b>                        | No data available        | None known              |
| <b>Bulk density</b>                            | No data available        |                         |
| <b>Liquid Density</b>                          | No data available        |                         |
| <b>Relative vapour density</b>                 | No data available        | None known              |
| <b>Particle characteristics</b>                |                          |                         |
| <b>Particle Size</b>                           | No information available |                         |
| <b>Particle Size Distribution</b>              | No information available |                         |

### 9.2. Other information

## SECTION 10: Stability and reactivity

### 10.1. Reactivity

|                   |                           |
|-------------------|---------------------------|
| <b>Reactivity</b> | No information available. |
|-------------------|---------------------------|

### 10.2. Chemical stability

|                  |                                 |
|------------------|---------------------------------|
| <b>Stability</b> | Stable under normal conditions. |
|------------------|---------------------------------|

### Explosion data

|                                         |       |
|-----------------------------------------|-------|
| <b>Sensitivity to mechanical impact</b> | None. |
| <b>Sensitivity to static discharge</b>  | None. |

**10.3. Possibility of hazardous reactions**

**Possibility of hazardous reactions** None under normal processing.

**10.4. Conditions to avoid**

**Conditions to avoid** None known based on information supplied.

**10.5. Incompatible materials**

**Incompatible materials** None known based on information supplied.

**10.6. Hazardous decomposition products**

**Hazardous decomposition products** None known based on information supplied.

**SECTION 11: Toxicological information****11.1. Information on toxicological effects****Information on likely routes of exposure****Product Information**

|                     |                                                                   |
|---------------------|-------------------------------------------------------------------|
| <b>Inhalation</b>   | Specific test data for the substance or mixture is not available. |
| <b>Eye contact</b>  | Specific test data for the substance or mixture is not available. |
| <b>Skin contact</b> | Specific test data for the substance or mixture is not available. |
| <b>Ingestion</b>    | Specific test data for the substance or mixture is not available. |

**Symptoms related to the physical, chemical and toxicological characteristics**

**Symptoms** No information available.

**Acute toxicity****Numerical measures of toxicity****Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Skin corrosion/irritation** No information available.

**Serious eye damage/eye irritation** No information available.

**Respiratory or skin sensitisation** No information available.

**Germ cell mutagenicity** No information available.

**Carcinogenicity** No information available.

**Reproductive toxicity** No information available.

STOT - single exposure No information available.

STOT - repeated exposure No information available.

Aspiration hazard No information available.

Other adverse effects No information available.

## SECTION 12: Ecological information

### 12.1. Toxicity

Ecotoxicity The environmental impact of this product has not been fully investigated.

### 12.2. Persistence and degradability

Persistence and degradability No information available.

### 12.3. Bioaccumulative potential

Bioaccumulation There is no data for this product.

### 12.4. Mobility in soil

Mobility in soil No information available.

### 12.5. Results of PBT and vPvB assessment

PBT and vPvB assessment No information available.

### 12.6. Endocrine disrupting properties

No information available.

## SECTION 13: Disposal considerations

### 13.1. Waste treatment methods

Waste from residues/unused products Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

Contaminated packaging Do not reuse empty containers.

## SECTION 14: Transport information

### IATA

|                                   |                |
|-----------------------------------|----------------|
| 14.1 UN number or ID number       | Not regulated  |
| 14.2 UN proper shipping name      | Not regulated  |
| 14.3 Transport hazard class(es)   | Not regulated  |
| 14.4 Packing group                | Not regulated  |
| 14.5 Environmental hazards        | Not applicable |
| 14.6 Special precautions for user |                |
| Special Provisions                | None           |

### IMDG



|                                                              |                          |
|--------------------------------------------------------------|--------------------------|
| 14.1 UN number or ID number                                  | Not regulated            |
| 14.2 UN proper shipping name                                 | Not regulated            |
| 14.3 Transport hazard class(es)                              | Not regulated            |
| 14.4 Packing group                                           | Not regulated            |
| 14.5 Environmental hazards                                   | Not applicable           |
| 14.6 Special precautions for user                            |                          |
| Special Provisions                                           | None                     |
| 14.7 Maritime transport in bulk according to IMO instruments | No information available |

**RID**

|                                   |                |
|-----------------------------------|----------------|
| 14.1 UN number or ID number       | Not regulated  |
| 14.2 UN proper shipping name      | Not regulated  |
| 14.3 Transport hazard class(es)   | Not regulated  |
| 14.4 Packing group                | Not regulated  |
| 14.5 Environmental hazards        | Not applicable |
| 14.6 Special precautions for user |                |
| Special Provisions                | None           |

**ADR**

|                                   |                |
|-----------------------------------|----------------|
| 14.1 UN number or ID number       | Not regulated  |
| 14.2 UN proper shipping name      | Not regulated  |
| 14.3 Transport hazard class(es)   | Not regulated  |
| 14.4 Packing group                | Not regulated  |
| 14.5 Environmental hazards        | Not applicable |
| 14.6 Special precautions for user |                |
| Special Provisions                | None           |

**SECTION 15: Regulatory information****15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture****Authorisations and/or restrictions on use:**

This product does not contain substances subject to authorisation (UK REACH - Annex XIV). This product does not contain substances subject to restriction (UK REACH - Annex XVII).

**Persistent Organic Pollutants**

Not applicable

**Export Notification requirements**

Not applicable

**Named dangerous substances per COMAH (SI 2015/483 as amended)**

Not applicable

**The Ozone-Depleting Substances Regulations 2015**

Not applicable

**The Biocidal Products Regulations 2001 (as amended)**

Not applicable

**The Water Environment (Water Framework Directive) (England and Wales) Regulations 2017 (as amended)**

Not applicable

**Poisons and Explosive Precursors**

Not applicable

**International Inventories**

Contact supplier for inventory compliance status

**15.2. Chemical safety assessment****Chemical Safety Report**

No information available

## SECTION 16: Other information

### Key or legend to abbreviations and acronyms used in the safety data sheet

#### Legend

SVHC: Substances of Very High Concern for Authorisation:

#### Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | Sk*  | Skin designation                 |
| +       | Sensitisers                 |      |                                  |

#### Classification procedure

|                                                                 |                    |
|-----------------------------------------------------------------|--------------------|
| Classification according to Regulation (EC) No. 1272/2008 [CLP] | Method Used        |
| Acute oral toxicity                                             | Calculation method |
| Acute dermal toxicity                                           | Calculation method |
| Acute inhalation toxicity - gas                                 | Calculation method |
| Acute inhalation toxicity - vapour                              | Calculation method |
| Acute inhalation toxicity - dust/mist                           | Calculation method |
| Skin corrosion/irritation                                       | Calculation method |
| Serious eye damage/eye irritation                               | Calculation method |
| Respiratory sensitisation                                       | Calculation method |
| Skin sensitisation                                              | Calculation method |
| Mutagenicity                                                    | Calculation method |
| Carcinogenicity                                                 | Calculation method |
| Reproductive toxicity                                           | Calculation method |
| STOT - single exposure                                          | Calculation method |
| STOT - repeated exposure                                        | Calculation method |
| Acute aquatic toxicity                                          | Calculation method |
| Chronic aquatic toxicity                                        | Calculation method |
| Aspiration hazard                                               | Calculation method |
| Ozone                                                           | Calculation method |

#### Key literature references and sources for data used to compile the SDS

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 European Chemicals Agency (ECHA) Committee for Risk Assessment (ECHA\_RAC)  
 European Chemicals Agency (ECHA) (ECHA\_API)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 U.S. National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
 Organisation for Economic Co-operation and Development Screening Information Data Set  
 World Health Organization

**Revision date** 12-Dec-2024

#### Revision Note

Significant changes throughout SDS. Review all sections

**This SDS complies with the requirements of UK REACH Regulations SI 2019/758 (as amended)**

**Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work**

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**

**UK SDS version information - XGHS**

UL release:

GHS Revision 7

2024 Q4



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
UK REACH Regulations (SI 2019/758 as amended)

Revision date 12-Dec-2024

Revision Number 1.1

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### 1.1. Product identifier

**Product Name** R3 - Negative control, 2.1 ml

**Catalogue Number(s)** 7253F

**Pure substance/mixture** Mixture

Contains 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

**Recommended use** Restricted to professional users In vitro diagnostic

**Uses advised against** No information available

### 1.3. Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories Ltd  
The Junction  
Station Road  
Watford, WD17 1ET  
UK

For further information, please contact

**Technical Service** 00800 00246 723  
Techsupport.UK@bio-rad.com

### 1.4. Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC UK: 44-870-8200418

## SECTION 2: Hazards identification

### 2.1. Classification of the substance or mixture

|                                                |                     |
|------------------------------------------------|---------------------|
| Skin sensitisation                             | Category 1 - (H317) |
| Hazardous to the aquatic environment - chronic | Category 3 - (H412) |

### 2.2. Label elements

Contains 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone



**Signal word**

Warning

**Hazard statements**

H317 - May cause an allergic skin reaction

H412 - Harmful to aquatic life with long lasting effects

**Precautionary statements**

P333 + P313 - If skin irritation or rash occurs: Get medical advice/attention

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water

P501 - Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

P280 - Wear protective gloves, protective clothing, eye protection and face protection

**2.3. Other hazards**

Contains animal source material. (Cattle).

**SECTION 3: Composition/information on ingredients****3.1 Substances**

Not applicable

**3.2 Mixtures**

| Chemical name                                                                                  | Weight-% | EC No (EU Index No)         | UK REACH registration number | Classification according to GB CLP (SI 2020/1567 as amended)                                                                                                                                                               | Specific concentration limit (SCL)                                                                                                                            | M-Factor | M-Factor (long-term) |
|------------------------------------------------------------------------------------------------|----------|-----------------------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------|
| 1,2,3-Propanetriol<br>56-81-5                                                                  | 20       | 200-289-5                   | -                            | -                                                                                                                                                                                                                          | -                                                                                                                                                             | -        | -                    |
| Hydrochloric acid<br>7647-01-0                                                                 | 0.7      | 231-595-7<br>(017-002-00-2) | -                            | Skin Corr. 1B<br>(H314)<br>Eye Irrit. 2<br>(H319)<br>STOT SE 3<br>(H335)                                                                                                                                                   | Eye Irrit. 2 ::<br>1%≤C<3%<br>Skin Corr. 1B ::<br>C≥5%<br>Skin Irrit. 2 ::<br>1%≤C<5%<br>STOT SE 3 ::<br>C≥10%                                                | -        | -                    |
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone<br>55965-84-9 | 0.003    | (613-167-00-5)              | -                            | Acute Tox. 3<br>(H301)<br>Acute Tox. 3<br>(H311)<br>Acute Tox. 3<br>(H331)<br>Skin Corr. 1B<br>(H314)<br>Eye Dam. 1<br>(H318)<br>Skin Sens. 1A<br>(H317)<br>(EUH071)<br>Aquatic Acute 1 (H400)<br>Aquatic Chronic 1 (H410) | Eye Irrit. 2 ::<br>0.06%≤C<0.6%<br>Skin Corr. 1C ::<br>C≥0.6%<br>Skin Irrit. 2 ::<br>0.06%≤C<0.6%<br>Skin Sens. 1A ::<br>C≥0.0015%<br>Eye Dam. 1 ::<br>C≥0.6% | 100      | 100                  |

**Full text of H- and EUH-phrases: see section 16**

This product does not contain candidate substances of very high concern at a concentration  $\geq 0.1\%$  (UK REACH Article 59)

## SECTION 4: First aid measures

### 4.1. Description of first aid measures

|                |                                                                                                                                   |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------|
| General advice | Show this safety data sheet to the doctor in attendance.                                                                          |
| Inhalation     | Remove to fresh air.                                                                                                              |
| Eye contact    | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a doctor.                 |
| Skin contact   | Wash with soap and water. May cause an allergic skin reaction. In the case of skin irritation or allergic reactions see a doctor. |
| Ingestion      | Rinse mouth.                                                                                                                      |

### 4.2. Most important symptoms and effects, both acute and delayed

|          |                                                                             |
|----------|-----------------------------------------------------------------------------|
| Symptoms | Itching. Rashes. Hives. Prolonged contact may cause redness and irritation. |
|----------|-----------------------------------------------------------------------------|

### 4.3. Indication of any immediate medical attention and special treatment needed

|                 |                                                                        |
|-----------------|------------------------------------------------------------------------|
| Note to doctors | May cause sensitisation in susceptible persons. Treat symptomatically. |
|-----------------|------------------------------------------------------------------------|

## SECTION 5: Firefighting measures

### 5.1. Extinguishing media

|                                |                                                                                                         |
|--------------------------------|---------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
| Large Fire                     | CAUTION: Use of water spray when fighting fire may be inefficient.                                      |
| Unsuitable extinguishing media | Do not scatter spilled material with high pressure water streams.                                       |

### 5.2. Special hazards arising from the substance or mixture

|                                            |                                                                               |
|--------------------------------------------|-------------------------------------------------------------------------------|
| Specific hazards arising from the chemical | Product is or contains a sensitiser. May cause sensitisation by skin contact. |
|--------------------------------------------|-------------------------------------------------------------------------------|

### 5.3. Advice for firefighters

|                                                                |                                                                                                                                    |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|

## SECTION 6: Accidental release measures

### 6.1. Personal precautions, protective equipment and emergency procedures

|                          |                                                                                                                                                                                                          |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Personal precautions     | Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. |
| For emergency responders | Use personal protection recommended in Section 8.                                                                                                                                                        |

### 6.2. Environmental precautions

**Environmental precautions** See Section 12 for additional Ecological Information.

### **6.3. Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Take up mechanically, placing in appropriate containers for disposal.

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

### **6.4. Reference to other sections**

**Reference to other sections** See section 8 for more information. See section 13 for more information.

## **SECTION 7: Handling and storage**

### **7.1. Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. In case of insufficient ventilation, wear suitable respiratory equipment. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash it before reuse.

**General hygiene considerations** Wash hands before breaks and immediately after handling the product.

### **7.2. Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place. Store according to product and label instructions.

### **7.3. Specific end use(s)**

**Risk Management Methods (RMM)** The information required is contained in this Safety Data Sheet.

## **SECTION 8: Exposure controls/personal protection**

### **8.1. Control parameters**

#### **Exposure Limits**

| Chemical name                  | United Kingdom                                                                     |
|--------------------------------|------------------------------------------------------------------------------------|
| 1,2,3-Propanetriol<br>56-81-5  | TWA: 10 mg/m <sup>3</sup><br>STEL: 30 mg/m <sup>3</sup>                            |
| Hydrochloric acid<br>7647-01-0 | TWA: 1 ppm<br>TWA: 2 mg/m <sup>3</sup><br>STEL: 5 ppm<br>STEL: 8 mg/m <sup>3</sup> |

**Biological occupational exposure limits** This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies.

**Derived No Effect Level (DNEL)** No information available.

**Predicted No Effect Concentration (PNEC)** No information available.

### **8.2. Exposure controls**

**Engineering controls** No information available.

**Personal protective equipment**

|                                 |                                                                                                                                                                          |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye/face protection</b>      | Wear safety glasses with side shields (or goggles).                                                                                                                      |
| <b>Hand protection</b>          | Wear suitable gloves.                                                                                                                                                    |
| <b>Skin and body protection</b> | Wear suitable protective clothing.                                                                                                                                       |
| <b>Respiratory protection</b>   | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |

**SECTION 9: Physical and chemical properties****9.1. Information on basic physical and chemical properties**

|                        |                          |
|------------------------|--------------------------|
| <b>Physical state</b>  | Liquid                   |
| <b>Appearance</b>      | Liquid                   |
| <b>Colour</b>          | light yellow             |
| <b>Odour</b>           | Odourless.               |
| <b>Odour threshold</b> | No information available |

| <b><u>Property</u></b>                         | <b><u>Values</u></b>     | <b><u>Remarks • Method</u></b> |
|------------------------------------------------|--------------------------|--------------------------------|
| <b>Melting point / freezing point</b>          | No data available        | None known                     |
| <b>Initial boiling point and boiling range</b> | No data available        | None known                     |
| <b>Flammability</b>                            | No data available        | None known                     |
| <b>Flammability Limit in Air</b>               |                          | None known                     |
| <b>Upper flammability or explosive limits</b>  | No data available        |                                |
| <b>Lower flammability or explosive limits</b>  | No data available        |                                |
| <b>Flash point</b>                             | No data available        | None known                     |
| <b>Autoignition temperature</b>                | No data available °C     |                                |
| <b>Decomposition temperature</b>               |                          | None known                     |
| <b>pH</b>                                      | 7.4                      | None known                     |
| <b>pH (as aqueous solution)</b>                | No data available        | None known                     |
| <b>Kinematic viscosity</b>                     | No data available        | None known                     |
| <b>Dynamic viscosity</b>                       | No data available        | None known                     |
| <b>Water solubility</b>                        | Miscible in water        |                                |
| <b>Solubility(ies)</b>                         | No data available        | None known                     |
| <b>Partition coefficient</b>                   | No data available        | None known                     |
| <b>Vapour pressure</b>                         | No data available        | None known                     |
| <b>Relative density</b>                        | No data available        | None known                     |
| <b>Bulk density</b>                            | No data available        |                                |
| <b>Liquid Density</b>                          | No data available        |                                |
| <b>Relative vapour density</b>                 | No data available        | None known                     |
| <b>Particle characteristics</b>                |                          |                                |
| <b>Particle Size</b>                           | No information available |                                |
| <b>Particle Size Distribution</b>              | No information available |                                |

**9.2. Other information****SECTION 10: Stability and reactivity****10.1. Reactivity**

|                   |                           |
|-------------------|---------------------------|
| <b>Reactivity</b> | No information available. |
|-------------------|---------------------------|

**10.2. Chemical stability**



**Stability** Stable under normal conditions.

**Explosion data**

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

**10.3. Possibility of hazardous reactions**

**Possibility of hazardous reactions** None under normal processing.

**10.4. Conditions to avoid**

**Conditions to avoid** None known based on information supplied.

**10.5. Incompatible materials**

**Incompatible materials** None known based on information supplied.

**10.6. Hazardous decomposition products**

**Hazardous decomposition products** None known based on information supplied.

## SECTION 11: Toxicological information

**11.1. Information on toxicological effects**

**Information on likely routes of exposure**

**Product Information**

**Inhalation** Specific test data for the substance or mixture is not available.

**Eye contact** Specific test data for the substance or mixture is not available.

**Skin contact** May cause sensitisation by skin contact. Specific test data for the substance or mixture is not available. Repeated or prolonged skin contact may cause allergic reactions with susceptible persons (based on components). Causes mild skin irritation.

**Ingestion** Specific test data for the substance or mixture is not available.

**Symptoms related to the physical, chemical and toxicological characteristics**

**Symptoms** Itching. Rashes. Hives. Prolonged contact may cause redness and irritation.

**Acute toxicity**

**Numerical measures of toxicity**

**Component Information**

| Chemical name                                                                    | Oral LD50               | Dermal LD50              | Inhalation LC50         |
|----------------------------------------------------------------------------------|-------------------------|--------------------------|-------------------------|
| 1,2,3-Propanetriol                                                               | = 27200 mg/kg ( Rat )   | > 10 g/kg ( Rabbit )     | > 5.85 mg/L ( Rat ) 4 h |
| Hydrochloric acid                                                                | 238 - 277 mg/kg ( Rat ) | > 5010 mg/kg ( Rabbit )  | = 1.68 mg/L ( Rat ) 1 h |
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone | = 53 mg/kg ( Rat )      | = 87.12 mg/kg ( Rabbit ) | -                       |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Skin corrosion/irritation** Classification based on data available for ingredients. Causes mild skin irritation.

|                                   |                                      |
|-----------------------------------|--------------------------------------|
| Serious eye damage/eye irritation | No information available.            |
| Respiratory or skin sensitisation | May cause an allergic skin reaction. |
| Germ cell mutagenicity            | No information available.            |
| Carcinogenicity                   | No information available.            |
| Reproductive toxicity             | No information available.            |
| STOT - single exposure            | No information available.            |
| STOT - repeated exposure          | No information available.            |
| Aspiration hazard                 | No information available.            |
| Other adverse effects             | No information available.            |

## SECTION 12: Ecological information

### 12.1. Toxicity

**Ecotoxicity** Harmful to aquatic life with long lasting effects.

| Chemical name      | Algae/aquatic plants | Fish                                         | Toxicity to microorganisms | Crustacea |
|--------------------|----------------------|----------------------------------------------|----------------------------|-----------|
| 1,2,3-Propanetriol | -                    | LC50: 51 - 57mL/L (96h, Oncorhynchus mykiss) | -                          | -         |

### 12.2. Persistence and degradability

**Persistence and degradability** No information available.

### 12.3. Bioaccumulative potential

**Bioaccumulation**

#### Component Information

| Chemical name                                                                    | Partition coefficient |
|----------------------------------------------------------------------------------|-----------------------|
| 1,2,3-Propanetriol                                                               | -1.75                 |
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone | 0.7                   |

### 12.4. Mobility in soil

**Mobility in soil** No information available.

### 12.5. Results of PBT and vPvB assessment

**PBT and vPvB assessment** No information available.

| Chemical name      | PBT and vPvB assessment         |
|--------------------|---------------------------------|
| 1,2,3-Propanetriol | The substance is not PBT / vPvB |

|                                                                                  |                                 |
|----------------------------------------------------------------------------------|---------------------------------|
| Hydrochloric acid                                                                | The substance is not PBT / vPvB |
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone | The substance is not PBT / vPvB |

**12.6. Endocrine disrupting properties**

No information available.

**SECTION 13: Disposal considerations****13.1. Waste treatment methods**

|                                            |                                                                                                                 |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Waste from residues/unused products</b> | Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. |
| <b>Contaminated packaging</b>              | Do not reuse empty containers.                                                                                  |

**SECTION 14: Transport information****IATA**

|                                          |                |
|------------------------------------------|----------------|
| <b>14.1 UN number or ID number</b>       | Not regulated  |
| <b>14.2 UN proper shipping name</b>      | Not regulated  |
| <b>14.3 Transport hazard class(es)</b>   | Not regulated  |
| <b>14.4 Packing group</b>                | Not regulated  |
| <b>14.5 Environmental hazards</b>        | Not applicable |
| <b>14.6 Special precautions for user</b> |                |
| Special Provisions                       | None           |

**IMDG**

|                                                                     |                          |
|---------------------------------------------------------------------|--------------------------|
| <b>14.1 UN number or ID number</b>                                  | Not regulated            |
| <b>14.2 UN proper shipping name</b>                                 | Not regulated            |
| <b>14.3 Transport hazard class(es)</b>                              | Not regulated            |
| <b>14.4 Packing group</b>                                           | Not regulated            |
| <b>14.5 Environmental hazards</b>                                   | Not applicable           |
| <b>14.6 Special precautions for user</b>                            |                          |
| Special Provisions                                                  | None                     |
| <b>14.7 Maritime transport in bulk according to IMO instruments</b> | No information available |

**RID**

|                                          |                |
|------------------------------------------|----------------|
| <b>14.1 UN number or ID number</b>       | Not regulated  |
| <b>14.2 UN proper shipping name</b>      | Not regulated  |
| <b>14.3 Transport hazard class(es)</b>   | Not regulated  |
| <b>14.4 Packing group</b>                | Not regulated  |
| <b>14.5 Environmental hazards</b>        | Not applicable |
| <b>14.6 Special precautions for user</b> |                |
| Special Provisions                       | None           |

**ADR**

|                                          |                |
|------------------------------------------|----------------|
| <b>14.1 UN number or ID number</b>       | Not regulated  |
| <b>14.2 UN proper shipping name</b>      | Not regulated  |
| <b>14.3 Transport hazard class(es)</b>   | Not regulated  |
| <b>14.4 Packing group</b>                | Not regulated  |
| <b>14.5 Environmental hazards</b>        | Not applicable |
| <b>14.6 Special precautions for user</b> |                |
| Special Provisions                       | None           |

**SECTION 15: Regulatory information****15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture**

**Authorisations and/or restrictions on use:**

This product does not contain substances subject to authorisation (UK REACH - Annex XIV). This product does not contain substances subject to restriction (UK REACH - Annex XVII).

**Persistent Organic Pollutants**

Not applicable

**Export Notification requirements**

Not applicable

**Named dangerous substances per COMAH (SI 2015/483 as amended)**

| Chemical name                 | Lower-tier requirements (tons) | Upper-tier requirements (tons) |
|-------------------------------|--------------------------------|--------------------------------|
| Hydrochloric acid - 7647-01-0 | 25                             | 250                            |

**The Ozone-Depleting Substances Regulations 2015**

Not applicable

**The Biocidal Products Regulations 2001 (as amended)**

| Chemical name                                                                                    | The Biocidal Products Regulations 2001 (as amended) |
|--------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| Hydrochloric acid - 7647-01-0                                                                    | PT2                                                 |
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone - 55965-84-9 | PT2<br>PT4<br>PT6<br>PT11<br>PT12<br>PT13           |

**The Water Environment (Water Framework Directive) (England and Wales) Regulations 2017 (as amended)**

Not applicable

**Poisons and Explosive Precursors**

| Chemical name     | Poisons and Explosive Precursors        |
|-------------------|-----------------------------------------|
| Hydrochloric acid | Explosive precursor, Regulated 10 % w/w |

**International Inventories**

Contact supplier for inventory compliance status

**15.2. Chemical safety assessment****Chemical Safety Report**

No information available

**SECTION 16: Other information****Key or legend to abbreviations and acronyms used in the safety data sheet****Full text of any hazard and/or precautionary statements referred to under Sections 2-15**

EUH071 - Corrosive to the respiratory tract

H301 - Toxic if swallowed

H311 - Toxic in contact with skin

H314 - Causes severe skin burns and eye damage

H317 - May cause an allergic skin reaction

H318 - Causes serious eye damage

H319 - Causes serious eye irritation

H331 - Toxic if inhaled

H335 - May cause respiratory irritation

H400 - Very toxic to aquatic life

H410 - Very toxic to aquatic life with long lasting effects

### Legend

SVHC: Substances of Very High Concern for Authorisation:

### Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | Sk*  | Skin designation                 |
| +       | Sensitisers                 |      |                                  |

### Classification procedure

|                                                                 |                    |
|-----------------------------------------------------------------|--------------------|
| Classification according to Regulation (EC) No. 1272/2008 [CLP] | Method Used        |
| Acute oral toxicity                                             | Calculation method |
| Acute dermal toxicity                                           | Calculation method |
| Acute inhalation toxicity - gas                                 | Calculation method |
| Acute inhalation toxicity - vapour                              | Calculation method |
| Acute inhalation toxicity - dust/mist                           | Calculation method |
| Skin corrosion/irritation                                       | Calculation method |
| Serious eye damage/eye irritation                               | Calculation method |
| Respiratory sensitisation                                       | Calculation method |
| Skin sensitisation                                              | Calculation method |
| Mutagenicity                                                    | Calculation method |
| Carcinogenicity                                                 | Calculation method |
| Reproductive toxicity                                           | Calculation method |
| STOT - single exposure                                          | Calculation method |
| STOT - repeated exposure                                        | Calculation method |
| Acute aquatic toxicity                                          | Calculation method |
| Chronic aquatic toxicity                                        | Calculation method |
| Aspiration hazard                                               | Calculation method |
| Ozone                                                           | Calculation method |

### Key literature references and sources for data used to compile the SDS

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 European Chemicals Agency (ECHA) Committee for Risk Assessment (ECHA\_RAC)  
 European Chemicals Agency (ECHA) (ECHA\_API)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 U.S. National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
 Organisation for Economic Co-operation and Development Screening Information Data Set  
 World Health Organization

Revision date 12-Dec-2024

### Revision Note

Significant changes throughout SDS. Review all sections

This SDS complies with the requirements of UK REACH Regulations SI 2019/758 (as amended)

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information

relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

End of Safety Data Sheet

UK SDS version information - XGHS

UL release:  
GHS Revision 7  
2024 Q4

Full text of any hazard and/or precautionary EUH071 - Corrosive to the respiratory tract H301 - Toxic if swallowed H311 - Toxic in contact with skin statements referred to under Sections 2-15 H314 - Causes severe skin burns and eye damage H317 - May cause an allergic skin reaction H318 - Causes serious eye damage H319 - Causes serious eye irritation H331 - Toxic if inhaled H335 - May cause respiratory irritation H400 - Very toxic to aquatic life H410 - Very toxic to aquatic life with long lasting effects

| Chemical name                                                                    | Classification according to GB CLP (SI 2020/1567 as amended)                                                                                                                                             | Specific concentration limit (SCL)                                                                                                                  |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Hydrochloric acid                                                                | Skin Corr. 1B (H314)<br>Eye Irrit. 2 (H319)<br>STOT SE 3 (H335)                                                                                                                                          | Eye Irrit. 2 :: 1%<=C<3%<br>Skin Corr. 1B :: C>=5%<br>Skin Irrit. 2 :: 1%<=C<5%<br>STOT SE 3 :: C>=10%                                              |
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone | Acute Tox. 3 (H301)<br>Acute Tox. 3 (H311)<br>Acute Tox. 3 (H331)<br>Skin Corr. 1B (H314)<br>Eye Dam. 1 (H318)<br>Skin Sens. 1A (H317)<br>(EUH071)<br>Aquatic Acute 1 (H400)<br>Aquatic Chronic 1 (H410) | Eye Irrit. 2 :: 0.06%<=C<0.6%<br>Skin Corr. 1C :: C>=0.6%<br>Skin Irrit. 2 :: 0.06%<=C<0.6%<br>Skin Sens. 1A :: C>=0.0015%<br>Eye Dam. 1 :: C>=0.6% |



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
UK REACH Regulations (SI 2019/758 as amended)

Revision date 12-Dec-2024

Revision Number 1.1

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### 1.1. Product identifier

**Product Name** R4 - Positive Control, 1.6 ml

**Catalogue Number(s)** 7253K

**Pure substance/mixture** Mixture

Contains 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

**Recommended use** Restricted to professional users In vitro diagnostic

**Uses advised against** No information available

### 1.3. Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories Ltd  
The Junction  
Station Road  
Watford, WD17 1ET  
UK

For further information, please contact

**Technical Service** 00800 00246 723  
Techsupport.UK@bio-rad.com

### 1.4. Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC UK: 44-870-8200418

## SECTION 2: Hazards identification

### 2.1. Classification of the substance or mixture

|                                                |                     |
|------------------------------------------------|---------------------|
| Skin sensitisation                             | Category 1 - (H317) |
| Hazardous to the aquatic environment - chronic | Category 3 - (H412) |

### 2.2. Label elements

Contains 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone



**Signal word**

Warning

**Hazard statements**

H317 - May cause an allergic skin reaction

H412 - Harmful to aquatic life with long lasting effects

**Precautionary statements**

P333 + P313 - If skin irritation or rash occurs: Get medical advice/attention

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water

P501 - Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

P280 - Wear protective gloves, protective clothing, eye protection and face protection

**2.3. Other hazards**

Contains animal source material. (Cattle).

Contains human source material and / or potentially infectious components

**SECTION 3: Composition/information on ingredients****3.1 Substances**

Not applicable

**3.2 Mixtures**

| Chemical name                                                                                  | Weight-% | EC No (EU Index No)         | UK REACH registration number | Classification according to GB CLP (SI 2020/1567 as amended)                                                                                                                                                               | Specific concentration limit (SCL)                                                                                                                            | M-Factor | M-Factor (long-term) |
|------------------------------------------------------------------------------------------------|----------|-----------------------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------|
| 1,2,3-Propanetriol<br>56-81-5                                                                  | 20       | 200-289-5                   | -                            | -                                                                                                                                                                                                                          | -                                                                                                                                                             | -        | -                    |
| Hydrochloric acid<br>7647-01-0                                                                 | 0.7      | 231-595-7<br>(017-002-00-2) | -                            | Skin Corr. 1B<br>(H314)<br>Eye Irrit. 2<br>(H319)<br>STOT SE 3<br>(H335)                                                                                                                                                   | Eye Irrit. 2 ::<br>1%≤C<3%<br>Skin Corr. 1B ::<br>C≥5%<br>Skin Irrit. 2 ::<br>1%≤C<5%<br>STOT SE 3 ::<br>C≥10%                                                | -        | -                    |
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone<br>55965-84-9 | 0.003    | (613-167-00-5)              | -                            | Acute Tox. 3<br>(H301)<br>Acute Tox. 3<br>(H311)<br>Acute Tox. 3<br>(H331)<br>Skin Corr. 1B<br>(H314)<br>Eye Dam. 1<br>(H318)<br>Skin Sens. 1A<br>(H317)<br>(EUH071)<br>Aquatic Acute 1 (H400)<br>Aquatic Chronic 1 (H410) | Eye Irrit. 2 ::<br>0.06%≤C<0.6%<br>Skin Corr. 1C ::<br>C≥0.6%<br>Skin Irrit. 2 ::<br>0.06%≤C<0.6%<br>Skin Sens. 1A ::<br>C≥0.0015%<br>Eye Dam. 1 ::<br>C≥0.6% | 100      | 100                  |

**Full text of H- and EUH-phrases: see section 16**



This product does not contain candidate substances of very high concern at a concentration  $\geq 0.1\%$  (UK REACH Article 59)

## SECTION 4: First aid measures

### 4.1. Description of first aid measures

|                |                                                                                                                                                                                    |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General advice | Show this safety data sheet to the doctor in attendance.                                                                                                                           |
| Inhalation     | Remove to fresh air.                                                                                                                                                               |
| Eye contact    | Contains human source material and / or potentially infectious components. Call a doctor. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. |
| Skin contact   | Wash with soap and water. May cause an allergic skin reaction. In the case of skin irritation or allergic reactions see a doctor.                                                  |
| Ingestion      | Contains human source material and / or potentially infectious components. Call a doctor.                                                                                          |

### 4.2. Most important symptoms and effects, both acute and delayed

|          |                                                                             |
|----------|-----------------------------------------------------------------------------|
| Symptoms | Itching. Rashes. Hives. Prolonged contact may cause redness and irritation. |
|----------|-----------------------------------------------------------------------------|

### 4.3. Indication of any immediate medical attention and special treatment needed

|                 |                                                                                                                                                   |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Note to doctors | May cause sensitisation in susceptible persons. Treat symptomatically. Contains human source material and / or potentially infectious components. |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------|

## SECTION 5: Firefighting measures

### 5.1. Extinguishing media

|                                |                                                                                                         |
|--------------------------------|---------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
| Large Fire                     | CAUTION: Use of water spray when fighting fire may be inefficient.                                      |
| Unsuitable extinguishing media | Do not scatter spilled material with high pressure water streams.                                       |

### 5.2. Special hazards arising from the substance or mixture

|                                            |                                                                               |
|--------------------------------------------|-------------------------------------------------------------------------------|
| Specific hazards arising from the chemical | Product is or contains a sensitiser. May cause sensitisation by skin contact. |
|--------------------------------------------|-------------------------------------------------------------------------------|

### 5.3. Advice for firefighters

|                                                                |                                                                                                                                    |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|

## SECTION 6: Accidental release measures

### 6.1. Personal precautions, protective equipment and emergency procedures

|                          |                                                                                                                                                                                                          |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Personal precautions     | Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. |
| For emergency responders | Use personal protection recommended in Section 8.                                                                                                                                                        |

**6.2. Environmental precautions**

**Environmental precautions** See Section 12 for additional Ecological Information.

**6.3. Methods and material for containment and cleaning up**

**Methods for containment** Do not allow into any sewer, on the ground or into any body of water.

**Methods for cleaning up** Use: Disinfectant. Clean contaminated surface thoroughly.

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

**6.4. Reference to other sections**

**Reference to other sections** See section 8 for more information. See section 13 for more information.

**SECTION 7: Handling and storage****7.1. Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. In case of insufficient ventilation, wear suitable respiratory equipment. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash it before reuse.

**General hygiene considerations** Follow universal and standard precautions for handling potentially infectious materials.

**7.2. Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place. Store according to product and label instructions.

**7.3. Specific end use(s)**

**Risk Management Methods (RMM)** The information required is contained in this Safety Data Sheet.

**SECTION 8: Exposure controls/personal protection****8.1. Control parameters****Exposure Limits**

| Chemical name                  | United Kingdom                                                                     |
|--------------------------------|------------------------------------------------------------------------------------|
| 1,2,3-Propanetriol<br>56-81-5  | TWA: 10 mg/m <sup>3</sup><br>STEL: 30 mg/m <sup>3</sup>                            |
| Hydrochloric acid<br>7647-01-0 | TWA: 1 ppm<br>TWA: 2 mg/m <sup>3</sup><br>STEL: 5 ppm<br>STEL: 8 mg/m <sup>3</sup> |

**Biological occupational exposure limits** This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies.

**Derived No Effect Level (DNEL)** No information available.

**Predicted No Effect Concentration (PNEC)** No information available.

**8.2. Exposure controls**

|                                             |                                                                                                                                                                          |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Engineering controls                        | No information available.                                                                                                                                                |
| <b><u>Personal protective equipment</u></b> |                                                                                                                                                                          |
| Eye/face protection                         | Wear safety glasses with side shields (or goggles).                                                                                                                      |
| Hand protection                             | Wear suitable gloves.                                                                                                                                                    |
| Skin and body protection                    | Wear suitable protective clothing.                                                                                                                                       |
| Respiratory protection                      | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |
| General hygiene considerations              | Follow universal and standard precautions for handling potentially infectious materials.                                                                                 |

## SECTION 9: Physical and chemical properties

### 9.1. Information on basic physical and chemical properties

|                 |                          |
|-----------------|--------------------------|
| Physical state  | Liquid                   |
| Appearance      | Liquid                   |
| Colour          | yellow                   |
| Odour           | Odourless.               |
| Odour threshold | No information available |

| <u>Property</u>                         | <u>Values</u>            | <u>Remarks • Method</u> |
|-----------------------------------------|--------------------------|-------------------------|
| Melting point / freezing point          | No data available        | None known              |
| Initial boiling point and boiling range | No data available        | None known              |
| Flammability                            | No data available        | None known              |
| Flammability Limit in Air               |                          | None known              |
| Upper flammability or explosive limits  | No data available        |                         |
| Lower flammability or explosive limits  | No data available        |                         |
| Flash point                             | No data available        | None known              |
| Autoignition temperature                | No data available °C     |                         |
| Decomposition temperature               |                          | None known              |
| pH                                      | 8                        | None known              |
| pH (as aqueous solution)                | No data available        | None known              |
| Kinematic viscosity                     | No data available        | None known              |
| Dynamic viscosity                       | No data available        | None known              |
| Water solubility                        | Miscible in water        |                         |
| Solubility(ies)                         | No data available        | None known              |
| Partition coefficient                   | No data available        | None known              |
| Vapour pressure                         | No data available        | None known              |
| Relative density                        | No data available        | None known              |
| Bulk density                            | No data available        |                         |
| Liquid Density                          | No data available        |                         |
| Relative vapour density                 | No data available        | None known              |
| Particle characteristics                |                          |                         |
| Particle Size                           | No information available |                         |
| Particle Size Distribution              | No information available |                         |

### 9.2. Other information

## SECTION 10: Stability and reactivity

### 10.1. Reactivity

|            |                           |
|------------|---------------------------|
| Reactivity | No information available. |
|------------|---------------------------|

**10.2. Chemical stability**

**Stability** Stable under normal conditions.

**Explosion data**

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

**10.3. Possibility of hazardous reactions**

**Possibility of hazardous reactions** None under normal processing.

**10.4. Conditions to avoid**

**Conditions to avoid** None known based on information supplied.

**10.5. Incompatible materials**

**Incompatible materials** None known based on information supplied.

**10.6. Hazardous decomposition products**

**Hazardous decomposition products** None known based on information supplied.

**SECTION 11: Toxicological information****11.1. Information on toxicological effects****Information on likely routes of exposure****Product Information**

**Inhalation** Specific test data for the substance or mixture is not available.

**Eye contact** Specific test data for the substance or mixture is not available.

**Skin contact** May cause sensitisation by skin contact. Specific test data for the substance or mixture is not available. Repeated or prolonged skin contact may cause allergic reactions with susceptible persons (based on components). Causes mild skin irritation.

**Ingestion** Specific test data for the substance or mixture is not available.

**Symptoms related to the physical, chemical and toxicological characteristics**

**Symptoms** Itching. Rashes. Hives. Prolonged contact may cause redness and irritation.

**Acute toxicity****Numerical measures of toxicity****Component Information**

| Chemical name                                                                    | Oral LD50               | Dermal LD50              | Inhalation LC50         |
|----------------------------------------------------------------------------------|-------------------------|--------------------------|-------------------------|
| 1,2,3-Propanetriol                                                               | = 27200 mg/kg ( Rat )   | > 10 g/kg ( Rabbit )     | > 5.85 mg/L ( Rat ) 4 h |
| Hydrochloric acid                                                                | 238 - 277 mg/kg ( Rat ) | > 5010 mg/kg ( Rabbit )  | = 1.68 mg/L ( Rat ) 1 h |
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone | = 53 mg/kg ( Rat )      | = 87.12 mg/kg ( Rabbit ) | -                       |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                                          |                                                                                      |
|------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Skin corrosion/irritation</b>         | Classification based on data available for ingredients. Causes mild skin irritation. |
| <b>Serious eye damage/eye irritation</b> | No information available.                                                            |
| <b>Respiratory or skin sensitisation</b> | May cause an allergic skin reaction.                                                 |
| <b>Germ cell mutagenicity</b>            | No information available.                                                            |
| <b>Carcinogenicity</b>                   | No information available.                                                            |
| <b>Reproductive toxicity</b>             | No information available.                                                            |
| <b>STOT - single exposure</b>            | No information available.                                                            |
| <b>STOT - repeated exposure</b>          | No information available.                                                            |
| <b>Aspiration hazard</b>                 | No information available.                                                            |
| <b>Other adverse effects</b>             | No information available.                                                            |

## SECTION 12: Ecological information

### 12.1. Toxicity

**Ecotoxicity** Harmful to aquatic life with long lasting effects.

| Chemical name      | Algae/aquatic plants | Fish                                                 | Toxicity to microorganisms | Crustacea |
|--------------------|----------------------|------------------------------------------------------|----------------------------|-----------|
| 1,2,3-Propanetriol | -                    | LC50: 51 - 57mL/L (96h, <i>Oncorhynchus mykiss</i> ) | -                          | -         |

### 12.2. Persistence and degradability

**Persistence and degradability** No information available.

### 12.3. Bioaccumulative potential

**Bioaccumulation**

#### Component Information

| Chemical name                                                                    | Partition coefficient |
|----------------------------------------------------------------------------------|-----------------------|
| 1,2,3-Propanetriol                                                               | -1.75                 |
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone | 0.7                   |

### 12.4. Mobility in soil

**Mobility in soil** No information available.

### 12.5. Results of PBT and vPvB assessment

**PBT and vPvB assessment** No information available.

| Chemical name                                                                       | PBT and vPvB assessment         |
|-------------------------------------------------------------------------------------|---------------------------------|
| 1,2,3-Propanetriol                                                                  | The substance is not PBT / vPvB |
| Hydrochloric acid                                                                   | The substance is not PBT / vPvB |
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone | The substance is not PBT / vPvB |

**12.6. Endocrine disrupting properties**

No information available.

**SECTION 13: Disposal considerations****13.1. Waste treatment methods**

|                                            |                                                                                                                 |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Waste from residues/unused products</b> | Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. |
| <b>Contaminated packaging</b>              | Do not reuse empty containers.                                                                                  |

**SECTION 14: Transport information****IATA**

|                                          |                |
|------------------------------------------|----------------|
| <b>14.1 UN number or ID number</b>       | Not regulated  |
| <b>14.2 UN proper shipping name</b>      | Not regulated  |
| <b>14.3 Transport hazard class(es)</b>   | Not regulated  |
| <b>14.4 Packing group</b>                | Not regulated  |
| <b>14.5 Environmental hazards</b>        | Not applicable |
| <b>14.6 Special precautions for user</b> |                |
| <b>Special Provisions</b>                | None           |

**IMDG**

|                                                                     |                          |
|---------------------------------------------------------------------|--------------------------|
| <b>14.1 UN number or ID number</b>                                  | Not regulated            |
| <b>14.2 UN proper shipping name</b>                                 | Not regulated            |
| <b>14.3 Transport hazard class(es)</b>                              | Not regulated            |
| <b>14.4 Packing group</b>                                           | Not regulated            |
| <b>14.5 Environmental hazards</b>                                   | Not applicable           |
| <b>14.6 Special precautions for user</b>                            |                          |
| <b>Special Provisions</b>                                           | None                     |
| <b>14.7 Maritime transport in bulk according to IMO instruments</b> | No information available |

**RID**

|                                          |                |
|------------------------------------------|----------------|
| <b>14.1 UN number or ID number</b>       | Not regulated  |
| <b>14.2 UN proper shipping name</b>      | Not regulated  |
| <b>14.3 Transport hazard class(es)</b>   | Not regulated  |
| <b>14.4 Packing group</b>                | Not regulated  |
| <b>14.5 Environmental hazards</b>        | Not applicable |
| <b>14.6 Special precautions for user</b> |                |
| <b>Special Provisions</b>                | None           |

**ADR**

|                                          |                |
|------------------------------------------|----------------|
| <b>14.1 UN number or ID number</b>       | Not regulated  |
| <b>14.2 UN proper shipping name</b>      | Not regulated  |
| <b>14.3 Transport hazard class(es)</b>   | Not regulated  |
| <b>14.4 Packing group</b>                | Not regulated  |
| <b>14.5 Environmental hazards</b>        | Not applicable |
| <b>14.6 Special precautions for user</b> |                |
| <b>Special Provisions</b>                | None           |

## SECTION 15: Regulatory information

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### Authorisations and/or restrictions on use:

This product does not contain substances subject to authorisation (UK REACH - Annex XIV). This product does not contain substances subject to restriction (UK REACH - Annex XVII).

#### Persistent Organic Pollutants

Not applicable

#### Export Notification requirements

Not applicable

#### Named dangerous substances per COMAH (SI 2015/483 as amended)

| Chemical name                 | Lower-tier requirements (tons) | Upper-tier requirements (tons) |
|-------------------------------|--------------------------------|--------------------------------|
| Hydrochloric acid - 7647-01-0 | 25                             | 250                            |

#### The Ozone-Depleting Substances Regulations 2015

Not applicable

#### The Biocidal Products Regulations 2001 (as amended)

| Chemical name                                                                                    | The Biocidal Products Regulations 2001 (as amended) |
|--------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| Hydrochloric acid - 7647-01-0                                                                    | PT2                                                 |
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone - 55965-84-9 | PT2<br>PT4<br>PT6<br>PT11<br>PT12<br>PT13           |

#### The Water Environment (Water Framework Directive) (England and Wales) Regulations 2017 (as amended)

Not applicable

#### Poisons and Explosive Precursors

| Chemical name     | Poisons and Explosive Precursors        |
|-------------------|-----------------------------------------|
| Hydrochloric acid | Explosive precursor, Regulated 10 % w/w |

#### International Inventories

Contact supplier for inventory compliance status

### 15.2. Chemical safety assessment

#### Chemical Safety Report

No information available

## SECTION 16: Other information

### Key or legend to abbreviations and acronyms used in the safety data sheet

#### Full text of any hazard and/or precautionary statements referred to under Sections 2-15

EUH071 - Corrosive to the respiratory tract

H301 - Toxic if swallowed

H311 - Toxic in contact with skin

H314 - Causes severe skin burns and eye damage

H317 - May cause an allergic skin reaction

H318 - Causes serious eye damage

H319 - Causes serious eye irritation

H331 - Toxic if inhaled  
H335 - May cause respiratory irritation  
H400 - Very toxic to aquatic life  
H410 - Very toxic to aquatic life with long lasting effects

**Legend**

SVHC: Substances of Very High Concern for Authorisation:

**Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | Sk*  | Skin designation                 |
| +       | Sensitisers                 |      |                                  |

**Classification procedure**

|                                                                 |                    |
|-----------------------------------------------------------------|--------------------|
| Classification according to Regulation (EC) No. 1272/2008 [CLP] | Method Used        |
| Acute oral toxicity                                             | Calculation method |
| Acute dermal toxicity                                           | Calculation method |
| Acute inhalation toxicity - gas                                 | Calculation method |
| Acute inhalation toxicity - vapour                              | Calculation method |
| Acute inhalation toxicity - dust/mist                           | Calculation method |
| Skin corrosion/irritation                                       | Calculation method |
| Serious eye damage/eye irritation                               | Calculation method |
| Respiratory sensitisation                                       | Calculation method |
| Skin sensitisation                                              | Calculation method |
| Mutagenicity                                                    | Calculation method |
| Carcinogenicity                                                 | Calculation method |
| Reproductive toxicity                                           | Calculation method |
| STOT - single exposure                                          | Calculation method |
| STOT - repeated exposure                                        | Calculation method |
| Acute aquatic toxicity                                          | Calculation method |
| Chronic aquatic toxicity                                        | Calculation method |
| Aspiration hazard                                               | Calculation method |
| Ozone                                                           | Calculation method |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
European Chemicals Agency (ECHA) Committee for Risk Assessment (ECHA\_RAC)  
European Chemicals Agency (ECHA) (ECHA\_API)  
Environmental Protection Agency  
Acute Exposure Guideline Level(s) (AEGL(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
National Institute of Technology and Evaluation (NITE)  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
U.S. National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
World Health Organization

Revision date 12-Dec-2024

**Revision Note**

Significant changes throughout SDS. Review all sections

**This SDS complies with the requirements of UK REACH Regulations SI 2019/758 (as amended)**

**Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work**

**Disclaimer**



The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

End of Safety Data Sheet

UK SDS version information - XGHS

UL release:  
GHS Revision 7  
2024 Q4

Full text of any hazard and/or precautionary EUH071 - Corrosive to the respiratory tract H301 - Toxic if swallowed H311 - Toxic in contact with skin statements referred to under Sections 2-15 H314 - Causes severe skin burns and eye damage H317 - May cause an allergic skin reaction H318 - Causes serious eye damage H319 - Causes serious eye irritation H331 - Toxic if inhaled H335 - May cause respiratory irritation H400 - Very toxic to aquatic life H410 - Very toxic to aquatic life with long lasting effects

| Chemical name                                                                    | Classification according to GB CLP (SI 2020/1567 as amended)                                                                                                                                             | Specific concentration limit (SCL)                                                                                                                  |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Hydrochloric acid                                                                | Skin Corr. 1B (H314)<br>Eye Irrit. 2 (H319)<br>STOT SE 3 (H335)                                                                                                                                          | Eye Irrit. 2 :: 1%<=C<3%<br>Skin Corr. 1B :: C>=5%<br>Skin Irrit. 2 :: 1%<=C<5%<br>STOT SE 3 :: C>=10%                                              |
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone | Acute Tox. 3 (H301)<br>Acute Tox. 3 (H311)<br>Acute Tox. 3 (H331)<br>Skin Corr. 1B (H314)<br>Eye Dam. 1 (H318)<br>Skin Sens. 1A (H317)<br>(EUH071)<br>Aquatic Acute 1 (H400)<br>Aquatic Chronic 1 (H410) | Eye Irrit. 2 :: 0.06%<=C<0.6%<br>Skin Corr. 1C :: C>=0.6%<br>Skin Irrit. 2 :: 0.06%<=C<0.6%<br>Skin Sens. 1A :: C>=0.0015%<br>Eye Dam. 1 :: C>=0.6% |



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
UK REACH Regulations (SI 2019/758 as amended)

Revision date 11-Dec-2024

Revision Number 1.1

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### 1.1. Product identifier

**Product Name** R6 - Conjugate, 8 ml

**Catalogue Number(s)** 7253L

**Pure substance/mixture** Mixture

Contains 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

**Recommended use** Restricted to professional users In vitro diagnostic

**Uses advised against** No information available

### 1.3. Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories Ltd  
The Junction  
Station Road  
Watford, WD17 1ET  
UK

For further information, please contact

**Technical Service** 00800 00246 723  
Techsupport.UK@bio-rad.com

### 1.4. Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC UK: 44-870-8200418

## SECTION 2: Hazards identification

### 2.1. Classification of the substance or mixture

|                           |                     |
|---------------------------|---------------------|
| <b>Skin sensitisation</b> | Category 1 - (H317) |
|---------------------------|---------------------|

### 2.2. Label elements

Contains 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone



**Signal word**

Warning

#### Hazard statements

H317 - May cause an allergic skin reaction

#### Precautionary statements

P333 + P313 - If skin irritation or rash occurs: Get medical advice/attention

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water

P501 - Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

P280 - Wear protective gloves, protective clothing, eye protection and face protection

### 2.3. Other hazards

Contains animal source material. (Cattle).

## SECTION 3: Composition/information on ingredients

### 3.1 Substances

Not applicable

### 3.2 Mixtures

| Chemical name                                                                                  | Weight-% | EC No (EU Index No) | UK REACH registration number | Classification according to GB CLP (SI 2020/1567 as amended)                                                                                                                                          | Specific concentration limit (SCL)                                                                                                             | M-Factor | M-Factor (long-term) |
|------------------------------------------------------------------------------------------------|----------|---------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone<br>55965-84-9 | 0.0015   | (613-167-00-5)      | -                            | Acute Tox. 3 (H301)<br>Acute Tox. 3 (H311)<br>Acute Tox. 3 (H331)<br>Skin Corr. 1B (H314)<br>Eye Dam. 1 (H318)<br>Skin Sens. 1A (H317) (EUH071)<br>Aquatic Acute 1 (H400)<br>Aquatic Chronic 1 (H410) | Eye Irrit. 2 :: 0.06%≤C<0.6%<br>Skin Corr. 1C :: C≥0.6%<br>Skin Irrit. 2 :: 0.06%≤C<0.6%<br>Skin Sens. 1A :: C≥0.0015%<br>Eye Dam. 1 :: C≥0.6% | 100      | 100                  |

#### Full text of H- and EUH-phrases: see section 16

This product does not contain candidate substances of very high concern at a concentration ≥ 0.1% (UK REACH Article 59)

## SECTION 4: First aid measures

### 4.1. Description of first aid measures

#### General advice

Show this safety data sheet to the doctor in attendance.

#### Inhalation

Remove to fresh air.

#### Eye contact

Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.

|                     |                                                                                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------|
|                     | Consult a doctor.                                                                                                                 |
| <b>Skin contact</b> | Wash with soap and water. May cause an allergic skin reaction. In the case of skin irritation or allergic reactions see a doctor. |
| <b>Ingestion</b>    | Rinse mouth.                                                                                                                      |

#### **4.2. Most important symptoms and effects, both acute and delayed**

|                 |                         |
|-----------------|-------------------------|
| <b>Symptoms</b> | Itching. Rashes. Hives. |
|-----------------|-------------------------|

#### **4.3. Indication of any immediate medical attention and special treatment needed**

|                        |                                                                        |
|------------------------|------------------------------------------------------------------------|
| <b>Note to doctors</b> | May cause sensitisation in susceptible persons. Treat symptomatically. |
|------------------------|------------------------------------------------------------------------|

### **SECTION 5: Firefighting measures**

#### **5.1. Extinguishing media**

|                                       |                                                                                                         |
|---------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
| <b>Large Fire</b>                     | CAUTION: Use of water spray when fighting fire may be inefficient.                                      |
| <b>Unsuitable extinguishing media</b> | Do not scatter spilled material with high pressure water streams.                                       |

#### **5.2. Special hazards arising from the substance or mixture**

|                                                   |                                                                               |
|---------------------------------------------------|-------------------------------------------------------------------------------|
| <b>Specific hazards arising from the chemical</b> | Product is or contains a sensitiser. May cause sensitisation by skin contact. |
|---------------------------------------------------|-------------------------------------------------------------------------------|

#### **5.3. Advice for firefighters**

|                                                                       |                                                                                                                                    |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Special protective equipment and precautions for fire-fighters</b> | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|

### **SECTION 6: Accidental release measures**

#### **6.1. Personal precautions, protective equipment and emergency procedures**

|                                 |                                                                                                                                                                                                          |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal precautions</b>     | Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. |
| <b>For emergency responders</b> | Use personal protection recommended in Section 8.                                                                                                                                                        |

#### **6.2. Environmental precautions**

|                                  |                                                       |
|----------------------------------|-------------------------------------------------------|
| <b>Environmental precautions</b> | See Section 12 for additional Ecological Information. |
|----------------------------------|-------------------------------------------------------|

#### **6.3. Methods and material for containment and cleaning up**

|                                        |                                                                                      |
|----------------------------------------|--------------------------------------------------------------------------------------|
| <b>Methods for containment</b>         | Prevent further leakage or spillage if safe to do so.                                |
| <b>Methods for cleaning up</b>         | Take up mechanically, placing in appropriate containers for disposal.                |
| <b>Prevention of secondary hazards</b> | Clean contaminated objects and areas thoroughly observing environmental regulations. |

#### **6.4. Reference to other sections**

|                                    |                                                                          |
|------------------------------------|--------------------------------------------------------------------------|
| <b>Reference to other sections</b> | See section 8 for more information. See section 13 for more information. |
|------------------------------------|--------------------------------------------------------------------------|

## SECTION 7: Handling and storage

### 7.1. Precautions for safe handling

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. In case of insufficient ventilation, wear suitable respiratory equipment. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash it before reuse.

**General hygiene considerations** Wash hands before breaks and immediately after handling the product.

### 7.2. Conditions for safe storage, including any incompatibilities

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place. Store according to product and label instructions.

### 7.3. Specific end use(s)

**Risk Management Methods (RMM)** The information required is contained in this Safety Data Sheet.

## SECTION 8: Exposure controls/personal protection

### 8.1. Control parameters

**Exposure Limits** This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Biological occupational exposure limits** This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies.

**Derived No Effect Level (DNEL)** No information available.

**Predicted No Effect Concentration (PNEC)** No information available.

### 8.2. Exposure controls

**Engineering controls** No information available.

### Personal protective equipment

**Eye/face protection** Wear safety glasses with side shields (or goggles).

**Hand protection** Wear suitable gloves.

**Skin and body protection** Wear suitable protective clothing.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

## SECTION 9: Physical and chemical properties

**9.1. Information on basic physical and chemical properties**

|                 |                          |
|-----------------|--------------------------|
| Physical state  | Liquid                   |
| Appearance      | Liquid                   |
| Colour          | red                      |
| Odour           | Odourless.               |
| Odour threshold | No information available |

| <u>Property</u>                         | <u>Values</u>            | <u>Remarks • Method</u> |
|-----------------------------------------|--------------------------|-------------------------|
| Melting point / freezing point          | No data available        | None known              |
| Initial boiling point and boiling range | No data available        | None known              |
| Flammability                            | No data available        | None known              |
| Flammability Limit in Air               |                          | None known              |
| Upper flammability or explosive limits  | No data available        |                         |
| Lower flammability or explosive limits  | No data available        |                         |
| Flash point                             | No data available        | None known              |
| Autoignition temperature                | No data available        | None known              |
| Decomposition temperature               |                          | None known              |
| pH                                      | 6.3                      | None known              |
| pH (as aqueous solution)                | No data available        | None known              |
| Kinematic viscosity                     | No data available        | None known              |
| Dynamic viscosity                       | No data available        | None known              |
| Water solubility                        | Miscible in water        |                         |
| Solubility(ies)                         | No data available        | None known              |
| Partition coefficient                   | No data available        | None known              |
| Vapour pressure                         | No data available        | None known              |
| Relative density                        | No data available        | None known              |
| Bulk density                            | No data available        |                         |
| Liquid Density                          | No data available        |                         |
| Relative vapour density                 | No data available        | None known              |
| Particle characteristics                |                          |                         |
| Particle Size                           | No information available |                         |
| Particle Size Distribution              | No information available |                         |

**9.2. Other information****SECTION 10: Stability and reactivity****10.1. Reactivity**

|            |                           |
|------------|---------------------------|
| Reactivity | No information available. |
|------------|---------------------------|

**10.2. Chemical stability**

|           |                                 |
|-----------|---------------------------------|
| Stability | Stable under normal conditions. |
|-----------|---------------------------------|

**Explosion data**

|                                  |       |
|----------------------------------|-------|
| Sensitivity to mechanical impact | None. |
| Sensitivity to static discharge  | None. |

**10.3. Possibility of hazardous reactions**

|                                    |                               |
|------------------------------------|-------------------------------|
| Possibility of hazardous reactions | None under normal processing. |
|------------------------------------|-------------------------------|

**10.4. Conditions to avoid**

|                     |                                           |
|---------------------|-------------------------------------------|
| Conditions to avoid | None known based on information supplied. |
|---------------------|-------------------------------------------|

**10.5. Incompatible materials**

|                        |                                           |
|------------------------|-------------------------------------------|
| Incompatible materials | None known based on information supplied. |
|------------------------|-------------------------------------------|

**10.6. Hazardous decomposition products**

**Hazardous decomposition products** None known based on information supplied.

## SECTION 11: Toxicological information

### 11.1. Information on toxicological effects

#### Information on likely routes of exposure

##### Product Information

|                     |                                                                                                                                                                                                                            |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>   | Specific test data for the substance or mixture is not available.                                                                                                                                                          |
| <b>Eye contact</b>  | Specific test data for the substance or mixture is not available.                                                                                                                                                          |
| <b>Skin contact</b> | May cause sensitisation by skin contact. Specific test data for the substance or mixture is not available. Repeated or prolonged skin contact may cause allergic reactions with susceptible persons (based on components). |
| <b>Ingestion</b>    | Specific test data for the substance or mixture is not available.                                                                                                                                                          |

#### Symptoms related to the physical, chemical and toxicological characteristics

**Symptoms** Itching. Rashes. Hives.

#### Acute toxicity

##### Numerical measures of toxicity

##### Component Information

| Chemical name                                                                    | Oral LD50          | Dermal LD50              | Inhalation LC50 |
|----------------------------------------------------------------------------------|--------------------|--------------------------|-----------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone | = 53 mg/kg ( Rat ) | = 87.12 mg/kg ( Rabbit ) | -               |

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                                          |                                      |
|------------------------------------------|--------------------------------------|
| <b>Skin corrosion/irritation</b>         | No information available.            |
| <b>Serious eye damage/eye irritation</b> | No information available.            |
| <b>Respiratory or skin sensitisation</b> | May cause an allergic skin reaction. |
| <b>Germ cell mutagenicity</b>            | No information available.            |
| <b>Carcinogenicity</b>                   | No information available.            |
| <b>Reproductive toxicity</b>             | No information available.            |
| <b>STOT - single exposure</b>            | No information available.            |
| <b>STOT - repeated exposure</b>          | No information available.            |

Aspiration hazard No information available.

Other adverse effects No information available.

## SECTION 12: Ecological information

### 12.1. Toxicity

Ecotoxicity The environmental impact of this product has not been fully investigated.

### 12.2. Persistence and degradability

Persistence and degradability No information available.

### 12.3. Bioaccumulative potential

Bioaccumulation

#### Component Information

| Chemical name                                                                    | Partition coefficient |
|----------------------------------------------------------------------------------|-----------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone | 0.7                   |

### 12.4. Mobility in soil

Mobility in soil No information available.

### 12.5. Results of PBT and vPvB assessment

PBT and vPvB assessment No information available.

| Chemical name                                                                    | PBT and vPvB assessment         |
|----------------------------------------------------------------------------------|---------------------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone | The substance is not PBT / vPvB |

### 12.6. Endocrine disrupting properties

No information available.

## SECTION 13: Disposal considerations

### 13.1. Waste treatment methods

Waste from residues/unused products Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

Contaminated packaging Do not reuse empty containers.

## SECTION 14: Transport information

### IATA

|                                   |                |
|-----------------------------------|----------------|
| 14.1 UN number or ID number       | Not regulated  |
| 14.2 UN proper shipping name      | Not regulated  |
| 14.3 Transport hazard class(es)   | Not regulated  |
| 14.4 Packing group                | Not regulated  |
| 14.5 Environmental hazards        | Not applicable |
| 14.6 Special precautions for user |                |
| Special Provisions                | None           |

### IMDG



|                                                                     |                          |
|---------------------------------------------------------------------|--------------------------|
| <b>14.1 UN number or ID number</b>                                  | Not regulated            |
| <b>14.2 UN proper shipping name</b>                                 | Not regulated            |
| <b>14.3 Transport hazard class(es)</b>                              | Not regulated            |
| <b>14.4 Packing group</b>                                           | Not regulated            |
| <b>14.5 Environmental hazards</b>                                   | Not applicable           |
| <b>14.6 Special precautions for user</b>                            |                          |
| <b>Special Provisions</b>                                           | None                     |
| <b>14.7 Maritime transport in bulk according to IMO instruments</b> | No information available |

**RID**

|                                          |                |
|------------------------------------------|----------------|
| <b>14.1 UN number or ID number</b>       | Not regulated  |
| <b>14.2 UN proper shipping name</b>      | Not regulated  |
| <b>14.3 Transport hazard class(es)</b>   | Not regulated  |
| <b>14.4 Packing group</b>                | Not regulated  |
| <b>14.5 Environmental hazards</b>        | Not applicable |
| <b>14.6 Special precautions for user</b> |                |
| <b>Special Provisions</b>                | None           |

**ADR**

|                                          |                |
|------------------------------------------|----------------|
| <b>14.1 UN number or ID number</b>       | Not regulated  |
| <b>14.2 UN proper shipping name</b>      | Not regulated  |
| <b>14.3 Transport hazard class(es)</b>   | Not regulated  |
| <b>14.4 Packing group</b>                | Not regulated  |
| <b>14.5 Environmental hazards</b>        | Not applicable |
| <b>14.6 Special precautions for user</b> |                |
| <b>Special Provisions</b>                | None           |

## SECTION 15: Regulatory information

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

**Authorisations and/or restrictions on use:**

This product does not contain substances subject to authorisation (UK REACH - Annex XIV). This product does not contain substances subject to restriction (UK REACH - Annex XVII).

**Persistent Organic Pollutants**

Not applicable

**Export Notification requirements**

Not applicable

**Named dangerous substances per COMAH (SI 2015/483 as amended)**

Not applicable

**The Ozone-Depleting Substances Regulations 2015**

Not applicable

**The Biocidal Products Regulations 2001 (as amended)**

| Chemical name                                                                                 | The Biocidal Products Regulations 2001 (as amended) |
|-----------------------------------------------------------------------------------------------|-----------------------------------------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone - 55965-84-9 | PT2<br>PT4<br>PT6<br>PT11<br>PT12<br>PT13           |

**The Water Environment (Water Framework Directive) (England and Wales) Regulations 2017 (as amended)**

Not applicable

**Poisons and Explosive Precursors**

Not applicable

**International Inventories**

Contact supplier for inventory compliance status

**15.2. Chemical safety assessment****Chemical Safety Report**

No information available

**SECTION 16: Other information****Key or legend to abbreviations and acronyms used in the safety data sheet****Full text of any hazard and/or precautionary statements referred to under Sections 2-15**

EUH071 - Corrosive to the respiratory tract

H301 - Toxic if swallowed

H311 - Toxic in contact with skin

H314 - Causes severe skin burns and eye damage

H317 - May cause an allergic skin reaction

H318 - Causes serious eye damage

H331 - Toxic if inhaled

H400 - Very toxic to aquatic life

H410 - Very toxic to aquatic life with long lasting effects

**Legend**

SVHC: Substances of Very High Concern for Authorisation:

**Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

TWA TWA (time-weighted average)

STEL

STEL (Short Term Exposure Limit)

Ceiling Maximum limit value

Sk\*

Skin designation

+ Sensitisers

**Classification procedure**

Classification according to Regulation (EC) No. 1272/2008 [CLP]

Method Used

Acute oral toxicity

Calculation method

Acute dermal toxicity

Calculation method

Acute inhalation toxicity - gas

Calculation method

Acute inhalation toxicity - vapour

Calculation method

Acute inhalation toxicity - dust/mist

Calculation method

Skin corrosion/irritation

Calculation method

Serious eye damage/eye irritation

Calculation method

Respiratory sensitisation

Calculation method

Skin sensitisation

Calculation method

Mutagenicity

Calculation method

Carcinogenicity

Calculation method

Reproductive toxicity

Calculation method

STOT - single exposure

Calculation method

STOT - repeated exposure

Calculation method

Acute aquatic toxicity

Calculation method

Chronic aquatic toxicity

Calculation method

Aspiration hazard

Calculation method

Ozone

Calculation method

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)

U.S. Environmental Protection Agency ChemView Database

European Food Safety Authority (EFSA)

European Chemicals Agency (ECHA) Committee for Risk Assessment (ECHA\_RAC)

European Chemicals Agency (ECHA) (ECHA\_API)

Environmental Protection Agency

Acute Exposure Guideline Level(s) (AEGL(s))

U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act

U.S. Environmental Protection Agency High Production Volume Chemicals

Food Research Journal

Hazardous Substance Database

International Uniform Chemical Information Database (IUCLID)

National Institute of Technology and Evaluation (NITE)  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
U.S. National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
World Health Organization

Revision date 11-Dec-2024

Revision Note Significant changes throughout SDS. Review all sections

**This SDS complies with the requirements of UK REACH Regulations SI 2019/758 (as amended)**

**Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work**

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**

#### UK SDS version information - XGHS

UL release:

GHS Revision 7

2024 Q4

Full text of any hazard and/or precautionary EUH071 - Corrosive to the respiratory tract H301 - Toxic if swallowed H311 - Toxic in contact with skin  
statements referred to under Sections 2-15 H314 - Causes severe skin burns and eye damage H317 - May cause an allergic skin reaction H318 -  
Causes serious eye damage H331 - Toxic if inhaled H400 - Very toxic to aquatic life H410 - Very toxic  
to aquatic life with long lasting effects

| Chemical name                                                                    | Classification according to GB CLP (SI 2020/1567 as amended)                                                                                                                                             | Specific concentration limit (SCL)                                                                                                             |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone | Acute Tox. 3 (H301)<br>Acute Tox. 3 (H311)<br>Acute Tox. 3 (H331)<br>Skin Corr. 1B (H314)<br>Eye Dam. 1 (H318)<br>Skin Sens. 1A (H317)<br>(EUH071)<br>Aquatic Acute 1 (H400)<br>Aquatic Chronic 1 (H410) | Eye Irrit. 2 :: 0.06%≤C<0.6%<br>Skin Corr. 1C :: C≥0.6%<br>Skin Irrit. 2 :: 0.06%≤C<0.6%<br>Skin Sens. 1A :: C≥0.0015%<br>Eye Dam. 1 :: C≥0.6% |



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
UK REACH Regulations (SI 2019/758 as amended)

Revision date 11-Dec-2024

Revision Number 1.2

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### 1.1. Product identifier

**Product Name** R7 - Conjugate 2, 14 ml

**Catalogue Number(s)** 7253H

**Pure substance/mixture** Mixture

Contains 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

**Recommended use** Restricted to professional users In vitro diagnostic

**Uses advised against** No information available

### 1.3. Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories Ltd  
The Junction  
Station Road  
Watford, WD17 1ET  
UK

For further information, please contact

**Technical Service** 00800 00246 723  
Techsupport.UK@bio-rad.com

### 1.4. Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC UK: 44-870-8200418

## SECTION 2: Hazards identification

### 2.1. Classification of the substance or mixture

|                           |                     |
|---------------------------|---------------------|
| <b>Skin sensitisation</b> | Category 1 - (H317) |
|---------------------------|---------------------|

### 2.2. Label elements

Contains 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone

**Signal word**

Warning

**Hazard statements**

H317 - May cause an allergic skin reaction

**Precautionary statements**

P333 + P313 - If skin irritation or rash occurs: Get medical advice/attention

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water

P501 - Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

P280 - Wear protective gloves, protective clothing, eye protection and face protection

**2.3. Other hazards**

Contains animal source material. (Cattle).

## SECTION 3: Composition/information on ingredients

**3.1 Substances**

Not applicable

**3.2 Mixtures**

| Chemical name                                                                                  | Weight-% | EC No (EU Index No) | UK REACH registration number | Classification according to GB CLP (SI 2020/1567 as amended)                                                                                                                                          | Specific concentration limit (SCL)                                                                                                             | M-Factor | M-Factor (long-term) |
|------------------------------------------------------------------------------------------------|----------|---------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone<br>55965-84-9 | 0.0015   | (613-167-00-5)      | -                            | Acute Tox. 3 (H301)<br>Acute Tox. 3 (H311)<br>Acute Tox. 3 (H331)<br>Skin Corr. 1B (H314)<br>Eye Dam. 1 (H318)<br>Skin Sens. 1A (H317) (EUH071)<br>Aquatic Acute 1 (H400)<br>Aquatic Chronic 1 (H410) | Eye Irrit. 2 :: 0.06%≤C<0.6%<br>Skin Corr. 1C :: C≥0.6%<br>Skin Irrit. 2 :: 0.06%≤C<0.6%<br>Skin Sens. 1A :: C≥0.0015%<br>Eye Dam. 1 :: C≥0.6% | 100      | 100                  |

**Full text of H- and EUH-phrases: see section 16**

This product does not contain candidate substances of very high concern at a concentration ≥ 0.1% (UK REACH Article 59)

**SECTION 4: First aid measures****4.1. Description of first aid measures**

|                |                                                                                                                                   |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------|
| General advice | Show this safety data sheet to the doctor in attendance.                                                                          |
| Inhalation     | Remove to fresh air.                                                                                                              |
| Eye contact    | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a doctor.                 |
| Skin contact   | Wash with soap and water. May cause an allergic skin reaction. In the case of skin irritation or allergic reactions see a doctor. |
| Ingestion      | Rinse mouth.                                                                                                                      |

**4.2. Most important symptoms and effects, both acute and delayed**

|          |                         |
|----------|-------------------------|
| Symptoms | Itching. Rashes. Hives. |
|----------|-------------------------|

**4.3. Indication of any immediate medical attention and special treatment needed**

|                 |                                                                        |
|-----------------|------------------------------------------------------------------------|
| Note to doctors | May cause sensitisation in susceptible persons. Treat symptomatically. |
|-----------------|------------------------------------------------------------------------|

**SECTION 5: Firefighting measures****5.1. Extinguishing media**

|                                |                                                                                                         |
|--------------------------------|---------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
| Large Fire                     | CAUTION: Use of water spray when fighting fire may be inefficient.                                      |
| Unsuitable extinguishing media | Do not scatter spilled material with high pressure water streams.                                       |

**5.2. Special hazards arising from the substance or mixture**

|                                            |                                                                               |
|--------------------------------------------|-------------------------------------------------------------------------------|
| Specific hazards arising from the chemical | Product is or contains a sensitizer. May cause sensitisation by skin contact. |
|--------------------------------------------|-------------------------------------------------------------------------------|

**5.3. Advice for firefighters**

|                                                                |                                                                                                                                    |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|

**SECTION 6: Accidental release measures****6.1. Personal precautions, protective equipment and emergency procedures**

|                          |                                                                                                                                                                                                          |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Personal precautions     | Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. |
| For emergency responders | Use personal protection recommended in Section 8.                                                                                                                                                        |

**6.2. Environmental precautions**

|                           |                                                       |
|---------------------------|-------------------------------------------------------|
| Environmental precautions | See Section 12 for additional Ecological Information. |
|---------------------------|-------------------------------------------------------|

**6.3. Methods and material for containment and cleaning up**

|                                        |                                                                                      |
|----------------------------------------|--------------------------------------------------------------------------------------|
| <b>Methods for containment</b>         | Prevent further leakage or spillage if safe to do so.                                |
| <b>Methods for cleaning up</b>         | Take up mechanically, placing in appropriate containers for disposal.                |
| <b>Prevention of secondary hazards</b> | Clean contaminated objects and areas thoroughly observing environmental regulations. |

#### **6.4. Reference to other sections**

|                                    |                                                                          |
|------------------------------------|--------------------------------------------------------------------------|
| <b>Reference to other sections</b> | See section 8 for more information. See section 13 for more information. |
|------------------------------------|--------------------------------------------------------------------------|

## **SECTION 7: Handling and storage**

### **7.1. Precautions for safe handling**

|                                       |                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Advice on safe handling</b>        | Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. In case of insufficient ventilation, wear suitable respiratory equipment. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash it before reuse. |
| <b>General hygiene considerations</b> | Wash hands before breaks and immediately after handling the product.                                                                                                                                                                                                                                                                  |

### **7.2. Conditions for safe storage, including any incompatibilities**

|                           |                                                                                                                             |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Storage Conditions</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Store according to product and label instructions. |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------|

### **7.3. Specific end use(s)**

|                                      |                                                                  |
|--------------------------------------|------------------------------------------------------------------|
| <b>Risk Management Methods (RMM)</b> | The information required is contained in this Safety Data Sheet. |
|--------------------------------------|------------------------------------------------------------------|

## **SECTION 8: Exposure controls/personal protection**

### **8.1. Control parameters**

|                                                 |                                                                                                                                                             |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Limits</b>                          | This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies. |
| <b>Biological occupational exposure limits</b>  | This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies.            |
| <b>Derived No Effect Level (DNEL)</b>           | No information available.                                                                                                                                   |
| <b>Predicted No Effect Concentration (PNEC)</b> | No information available.                                                                                                                                   |

### **8.2. Exposure controls**

|                             |                           |
|-----------------------------|---------------------------|
| <b>Engineering controls</b> | No information available. |
|-----------------------------|---------------------------|

#### **Personal protective equipment**

|                                 |                                                     |
|---------------------------------|-----------------------------------------------------|
| <b>Eye/face protection</b>      | Wear safety glasses with side shields (or goggles). |
| <b>Hand protection</b>          | Wear suitable gloves.                               |
| <b>Skin and body protection</b> | Wear suitable protective clothing.                  |

**Respiratory protection**

No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

## SECTION 9: Physical and chemical properties

**9.1. Information on basic physical and chemical properties**

|                        |                          |
|------------------------|--------------------------|
| <b>Physical state</b>  | Liquid                   |
| <b>Appearance</b>      | aqueous solution         |
| <b>Colour</b>          | red                      |
| <b>Odour</b>           | Odourless.               |
| <b>Odour threshold</b> | No information available |

| <b>Property</b>                                | <b>Values</b>            | <b>Remarks • Method</b> |
|------------------------------------------------|--------------------------|-------------------------|
| <b>Melting point / freezing point</b>          | No data available        | None known              |
| <b>Initial boiling point and boiling range</b> | No data available        | None known              |
| <b>Flammability</b>                            | No data available        | None known              |
| <b>Flammability Limit in Air</b>               |                          | None known              |
| <b>Upper flammability or explosive limits</b>  | No data available        |                         |
| <b>Lower flammability or explosive limits</b>  | No data available        |                         |
| <b>Flash point</b>                             | No data available        | None known              |
| <b>Autoignition temperature</b>                | No data available        | None known              |
| <b>Decomposition temperature</b>               |                          | None known              |
| <b>pH</b>                                      | 6.3                      |                         |
| <b>pH (as aqueous solution)</b>                | No data available        | None known              |
| <b>Kinematic viscosity</b>                     | No data available        | None known              |
| <b>Dynamic viscosity</b>                       | No data available        | None known              |
| <b>Water solubility</b>                        | Miscible in water        |                         |
| <b>Solubility(ies)</b>                         | No data available        | None known              |
| <b>Partition coefficient</b>                   | No data available        | None known              |
| <b>Vapour pressure</b>                         | No data available        | None known              |
| <b>Relative density</b>                        | No data available        | None known              |
| <b>Bulk density</b>                            | No data available        |                         |
| <b>Liquid Density</b>                          | No data available        |                         |
| <b>Relative vapour density</b>                 | No data available        | None known              |
| <b>Particle characteristics</b>                |                          |                         |
| <b>Particle Size</b>                           | No information available |                         |
| <b>Particle Size Distribution</b>              | No information available |                         |

**9.2. Other information**

## SECTION 10: Stability and reactivity

**10.1. Reactivity**

|                   |                           |
|-------------------|---------------------------|
| <b>Reactivity</b> | No information available. |
|-------------------|---------------------------|

**10.2. Chemical stability**

|                  |                                 |
|------------------|---------------------------------|
| <b>Stability</b> | Stable under normal conditions. |
|------------------|---------------------------------|

**Explosion data**

|                                         |       |
|-----------------------------------------|-------|
| <b>Sensitivity to mechanical impact</b> | None. |
| <b>Sensitivity to static discharge</b>  | None. |

**10.3. Possibility of hazardous reactions**

|                                           |                               |
|-------------------------------------------|-------------------------------|
| <b>Possibility of hazardous reactions</b> | None under normal processing. |
|-------------------------------------------|-------------------------------|



**10.4. Conditions to avoid**

**Conditions to avoid** None known based on information supplied.

**10.5. Incompatible materials**

**Incompatible materials** None known based on information supplied.

**10.6. Hazardous decomposition products**

**Hazardous decomposition products** None known based on information supplied.

**SECTION 11: Toxicological information****11.1. Information on toxicological effects****Information on likely routes of exposure****Product Information**

|                     |                                                                                                                                                                                                                            |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>   | Specific test data for the substance or mixture is not available.                                                                                                                                                          |
| <b>Eye contact</b>  | Specific test data for the substance or mixture is not available.                                                                                                                                                          |
| <b>Skin contact</b> | May cause sensitisation by skin contact. Specific test data for the substance or mixture is not available. Repeated or prolonged skin contact may cause allergic reactions with susceptible persons (based on components). |
| <b>Ingestion</b>    | Specific test data for the substance or mixture is not available.                                                                                                                                                          |

**Symptoms related to the physical, chemical and toxicological characteristics**

**Symptoms** Itching. Rashes. Hives.

**Acute toxicity****Numerical measures of toxicity****Component Information**

| Chemical name                                                                    | Oral LD50          | Dermal LD50              | Inhalation LC50 |
|----------------------------------------------------------------------------------|--------------------|--------------------------|-----------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone | = 53 mg/kg ( Rat ) | = 87.12 mg/kg ( Rabbit ) | -               |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                                          |                                      |
|------------------------------------------|--------------------------------------|
| <b>Skin corrosion/irritation</b>         | No information available.            |
| <b>Serious eye damage/eye irritation</b> | No information available.            |
| <b>Respiratory or skin sensitisation</b> | May cause an allergic skin reaction. |
| <b>Germ cell mutagenicity</b>            | No information available.            |
| <b>Carcinogenicity</b>                   | No information available.            |

|                          |                           |
|--------------------------|---------------------------|
| Reproductive toxicity    | No information available. |
| STOT - single exposure   | No information available. |
| STOT - repeated exposure | No information available. |
| Aspiration hazard        | No information available. |
| Other adverse effects    | No information available. |

## SECTION 12: Ecological information

### 12.1. Toxicity

|             |                                                                           |
|-------------|---------------------------------------------------------------------------|
| Ecotoxicity | The environmental impact of this product has not been fully investigated. |
|-------------|---------------------------------------------------------------------------|

### 12.2. Persistence and degradability

|                               |                           |
|-------------------------------|---------------------------|
| Persistence and degradability | No information available. |
|-------------------------------|---------------------------|

### 12.3. Bioaccumulative potential

#### Bioaccumulation

#### Component Information

| Chemical name                                                                       | Partition coefficient |
|-------------------------------------------------------------------------------------|-----------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone | 0.7                   |

### 12.4. Mobility in soil

|                  |                           |
|------------------|---------------------------|
| Mobility in soil | No information available. |
|------------------|---------------------------|

### 12.5. Results of PBT and vPvB assessment

|                         |                           |
|-------------------------|---------------------------|
| PBT and vPvB assessment | No information available. |
|-------------------------|---------------------------|

| Chemical name                                                                       | PBT and vPvB assessment         |
|-------------------------------------------------------------------------------------|---------------------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone | The substance is not PBT / vPvB |

### 12.6. Endocrine disrupting properties

No information available.

## SECTION 13: Disposal considerations

### 13.1. Waste treatment methods

|                                     |                                                                                                                 |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Waste from residues/unused products | Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. |
| Contaminated packaging              | Do not reuse empty containers.                                                                                  |

## SECTION 14: Transport information

**IATA**

|                                   |                |
|-----------------------------------|----------------|
| 14.1 UN number or ID number       | Not regulated  |
| 14.2 UN proper shipping name      | Not regulated  |
| 14.3 Transport hazard class(es)   | Not regulated  |
| 14.4 Packing group                | Not regulated  |
| 14.5 Environmental hazards        | Not applicable |
| 14.6 Special precautions for user |                |
| Special Provisions                | None           |

**IMDG**

|                                                              |                          |
|--------------------------------------------------------------|--------------------------|
| 14.1 UN number or ID number                                  | Not regulated            |
| 14.2 UN proper shipping name                                 | Not regulated            |
| 14.3 Transport hazard class(es)                              | Not regulated            |
| 14.4 Packing group                                           | Not regulated            |
| 14.5 Environmental hazards                                   | Not applicable           |
| 14.6 Special precautions for user                            |                          |
| Special Provisions                                           | None                     |
| 14.7 Maritime transport in bulk according to IMO instruments | No information available |

**RID**

|                                   |                |
|-----------------------------------|----------------|
| 14.1 UN number or ID number       | Not regulated  |
| 14.2 UN proper shipping name      | Not regulated  |
| 14.3 Transport hazard class(es)   | Not regulated  |
| 14.4 Packing group                | Not regulated  |
| 14.5 Environmental hazards        | Not applicable |
| 14.6 Special precautions for user |                |
| Special Provisions                | None           |

**ADR**

|                                   |                |
|-----------------------------------|----------------|
| 14.1 UN number or ID number       | Not regulated  |
| 14.2 UN proper shipping name      | Not regulated  |
| 14.3 Transport hazard class(es)   | Not regulated  |
| 14.4 Packing group                | Not regulated  |
| 14.5 Environmental hazards        | Not applicable |
| 14.6 Special precautions for user |                |
| Special Provisions                | None           |

## SECTION 15: Regulatory information

**15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture****Authorisations and/or restrictions on use:**

This product does not contain substances subject to authorisation (UK REACH - Annex XIV). This product does not contain substances subject to restriction (UK REACH - Annex XVII).

**Persistent Organic Pollutants**

Not applicable

**Export Notification requirements**

Not applicable

**Named dangerous substances per COMAH (SI 2015/483 as amended)**

Not applicable

**The Ozone-Depleting Substances Regulations 2015**

Not applicable

**The Biocidal Products Regulations 2001 (as amended)**

| Chemical name                                                                                    | The Biocidal Products Regulations 2001 (as amended) |
|--------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone - 55965-84-9 | PT2<br>PT4                                          |

|  |                             |
|--|-----------------------------|
|  | PT6<br>PT11<br>PT12<br>PT13 |
|--|-----------------------------|

**The Water Environment (Water Framework Directive) (England and Wales) Regulations 2017 (as amended)**

Not applicable

**Poisons and Explosive Precursors**

Not applicable

**International Inventories**

Contact supplier for inventory compliance status

**15.2. Chemical safety assessment****Chemical Safety Report**

No information available

**SECTION 16: Other information****Key or legend to abbreviations and acronyms used in the safety data sheet****Full text of any hazard and/or precautionary statements referred to under Sections 2-15**

EUH071 - Corrosive to the respiratory tract

H301 - Toxic if swallowed

H311 - Toxic in contact with skin

H314 - Causes severe skin burns and eye damage

H317 - May cause an allergic skin reaction

H318 - Causes serious eye damage

H331 - Toxic if inhaled

H400 - Very toxic to aquatic life

H410 - Very toxic to aquatic life with long lasting effects

**Legend**

SVHC: Substances of Very High Concern for Authorisation:

**Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

TWA TWA (time-weighted average)

Ceiling Maximum limit value

+ Sensitisers

STEL

Sk\*

STEL (Short Term Exposure Limit)

Skin designation

**Classification procedure**

Classification according to Regulation (EC) No. 1272/2008 [CLP]

Acute oral toxicity

Acute dermal toxicity

Acute inhalation toxicity - gas

Acute inhalation toxicity - vapour

Acute inhalation toxicity - dust/mist

Skin corrosion/irritation

Serious eye damage/eye irritation

Respiratory sensitisation

Skin sensitisation

Mutagenicity

Carcinogenicity

Reproductive toxicity

STOT - single exposure

STOT - repeated exposure

Acute aquatic toxicity

Chronic aquatic toxicity

Aspiration hazard

Ozone

Method Used

Calculation method

Calculation method

Calculation method

Calculation method

Calculation method

Calculation method

Calculation method

Calculation method

Calculation method

Calculation method

Calculation method

Calculation method

Calculation method

Calculation method

Calculation method

Calculation method

Calculation method

Calculation method

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)

U.S. Environmental Protection Agency ChemView Database

European Food Safety Authority (EFSA)  
European Chemicals Agency (ECHA) Committee for Risk Assessment (ECHA\_RAC)  
European Chemicals Agency (ECHA) (ECHA\_API)  
Environmental Protection Agency  
Acute Exposure Guideline Level(s) (AEGL(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
National Institute of Technology and Evaluation (NITE)  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
U.S. National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
World Health Organization

**Revision date** 11-Dec-2024

**Revision Note** Significant changes throughout SDS. Review all sections

**This SDS complies with the requirements of UK REACH Regulations SI 2019/758 (as amended)**

**Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work**

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**

#### UK SDS version information - XGHS

UL release:

GHS Revision 7

2024 Q4

Full text of any hazard and/or precautionary EUH071 - Corrosive to the respiratory tract H301 - Toxic if swallowed H311 - Toxic in contact with skin statements referred to under Sections 2-15 H314 - Causes severe skin burns and eye damage H317 - May cause an allergic skin reaction H318 - Causes serious eye damage H331 - Toxic if inhaled H400 - Very toxic to aquatic life H410 - Very toxic to aquatic life with long lasting effects

| Chemical name                                                                    | Classification according to GB CLP (SI 2020/1567 as amended)                                                                                                                                             | Specific concentration limit (SCL)                                                                                                             |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone | Acute Tox. 3 (H301)<br>Acute Tox. 3 (H311)<br>Acute Tox. 3 (H331)<br>Skin Corr. 1B (H314)<br>Eye Dam. 1 (H318)<br>Skin Sens. 1A (H317)<br>(EUH071)<br>Aquatic Acute 1 (H400)<br>Aquatic Chronic 1 (H410) | Eye Irrit. 2 :: 0.06%≤C<0.6%<br>Skin Corr. 1C :: C≥0.6%<br>Skin Irrit. 2 :: 0.06%≤C<0.6%<br>Skin Sens. 1A :: C≥0.0015%<br>Eye Dam. 1 :: C≥0.6% |